



Clinical use of neuromuscular blocking agents in anesthesia

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INTRODUCTION

Neuromuscular blocking agents (NMBAs) are usually administered during anesthesia to facilitate endotracheal intubation and/or to improve surgical conditions. NMBAs may decrease the incidence of hoarseness and vocal cord injuries during intubation, and can facilitate mechanical ventilation in patients with poor lung compliance [1-5].

This topic will discuss the clinical use and pharmacology of the commonly used NMBAs and reversal of neuromuscular block. Monitoring neuromuscular blockade is discussed separately. (See "[Monitoring neuromuscular blockade](#)".)

SELECTION OF NEUROMUSCULAR BLOCKING AGENTS

The selection of the appropriate NMBA depends on the clinical application and patient factors. For patients without contraindications to [succinylcholine](#) (eg, hyperkalemia, burns, stroke, susceptibility to malignant hyperthermia), options include the following (see '[Adverse effects of succinylcholine](#)' below):

- For endotracheal intubation for short procedures (<30 minutes), or if neuromonitoring will be used soon after intubation, a short duration of neuromuscular block is required. Options for endotracheal intubation include [succinylcholine](#), intubation without an NMBA (eg, high dose [remifentanyl](#) intubation, although this technique is associated with more trauma), or [rocuronium](#) or [vecuronium](#) if [sugammadex](#) is available for rapid reversal of block. (See '[Sugammadex](#)' below.)
- For endotracheal intubation for longer procedures (≥30 minutes), [succinylcholine](#) or any of the short or intermediate acting nondepolarizing NMBAs can be used for endotracheal intubation ([rocuronium](#), [vecuronium](#), mivacurium, [atracurium](#), or [cisatracurium](#)). The choice among these agents depends on availability, cost, and patient factors that affect metabolism. Pancuronium (no longer available in North America, Europe, and most areas around the world) is a rarely used, long acting alternative and should be avoided. (See '[Nondepolarizing neuromuscular blocking agents](#)' below.)

Intraoperative relaxation can be maintained as necessary with additional doses of nondepolarizing NMBA.

- [Succinylcholine](#) provides the most reliable and fastest intubating conditions, and is therefore the preferred NMBA for rapid sequence induction and intubation (RSII). Alternatives to succinylcholine for RSII include high dose [rocuronium](#) (1.2 mg/kg which is a 4 X ED95 dose), or avoidance of NMBAs with a high dose [remifentanyl](#) intubation. (See "[Rapid sequence induction and intubation \(RSII\) for anesthesia](#)", section on '[Neuromuscular blocking agents \(NMBAs\)](#)'.)

FACTORS THAT AFFECT THE RESPONSE TO NEUROMUSCULAR BLOCKING AGENTS

Patient factors — A number of patient factors can affect the response to NMBAs or require modification of the dose or interval of administration.

- **Neuromuscular disease** – NMBAs should be avoided or doses should be modified in patients with some neuromuscular diseases ([table 1](#)).
- [Succinylcholine](#) may cause life threatening hyperkalemia, and should be avoided, in conditions associated with denervation and associated upregulation of nicotinic acetylcholine receptors (nAChRs). (See '[Adverse effects of succinylcholine](#)' below.)
- Sensitivity to nondepolarizing NMBAs may be increased or variable in patients with neuromuscular diseases, and administration should be titrated, using careful monitoring.
- Patients with myasthenia gravis are resistant to [succinylcholine](#), but extremely sensitive to nondepolarizing NMBAs. The use of NMBAs in patients with myasthenia gravis is discussed separately. (See "[Anesthesia for the patient with myasthenia gravis or Lambert-Eton](#)".)

[myasthenic syndrome](#)", [section on 'Neuromuscular blocking agents'](#).)

- **Succinylcholine** is contraindicated in patients with neuromuscular diseases associated with susceptibility to malignant hyperthermia ([table 2](#)). (See '[Adverse effects of succinylcholine](#)' below and "[Malignant hyperthermia susceptibility: Evaluation](#)".)
- **Burns** – Burn injury upregulates extrajunctional nAChRs beginning within 24 hours of injury. As a result, administration of [succinylcholine](#) after 48 hours, and for up to a year after a major burn, can cause severe hyperkalemia and life threatening arrhythmias, and should be avoided. Upregulation of nAChRs also causes resistance to nondepolarizing NMBAs and reduces duration of action. (See "[Anesthesia for patients with acute burn injuries](#)", [section on 'Use of succinylcholine'](#).)
- **Extremes of age** – In older adult patients, the effects of steroidal NMBAs are prolonged due to decreased volume of distribution, changes in circulatory physiology, decreased regional renal and hepatic blood flow, and anatomic changes at the neuromuscular junction (NMJ). As such, smaller doses and less frequent dosing intervals may be warranted. The use of objective monitoring is particularly useful in this patient population ([table 3](#) and [table 4](#)).

Administration of NMBAs in infants and children is discussed separately. (See "[General anesthesia in neonates and children: Agents and techniques](#)", [section on 'Neuromuscular blocking agents'](#).)

- **Obesity** – The dose of [succinylcholine](#) should be based on total body weight for optimal intubating conditions [6]. Studies of the effects of obesity on the pharmacodynamics of various nondepolarizing NMBAs are conflicting, and the optimal method for calculating doses in patients with severe obesity is unclear [7,8]. We calculate doses for nondepolarizing NMBAs based on ideal body weight plus 10 percent, and use objective neuromuscular monitoring to minimize the risk of prolonged paralysis [9].
- **Hepatic and kidney disease** – Benzylisoquinolinium NMBAs are preferred for patients with hepatic or kidney dysfunction, since clearance of these drugs is independent of organ function. The duration of action of the steroidal NMBAs may be prolonged in these patients because of decreased clearance ([table 5](#)).

[Succinylcholine](#) may be used for patients with kidney disease, as long as serum potassium is normal. In these patients, the transient rise in potassium associated with succinylcholine is similar to patients without kidney disease [10]. (See '[Adverse effects of succinylcholine](#)' below.)

• Physiologic derangements

- **Hypothermia** – Hypothermia causes a prolonged response to nondepolarizing NMBAs as lower temperatures can affect excretion, volume of distribution, interactions with postjunctional nAChRs, and the pH at the NMJ [11,12]. This response is proportional to the degree of hypothermia but becomes more pronounced when muscle temperature is lower than 35.2°C [13]. It is important to note that neuromuscular monitoring is still reliable in this setting and can help guide neuromuscular blockade management in the hypothermic patient.

• Electrolyte disturbances

- Hypermagnesemia causes muscle relaxation, potentiates the effects of NMBAs, and can prolong the duration of action of [rocuronium](#), [cisatracurium](#), and [vecuronium](#). Hypermagnesemia is usually iatrogenic. As an example, magnesium is routinely administered in the intrapartum and postpartum period for patients with preeclampsia to prevent seizures. If these patients require general anesthesia, nondepolarizing NMBAs are rarely necessary and should be avoided. Since magnesium does not potentiate the effects of [succinylcholine](#), usual rapid sequence induction and intubation doses (succinylcholine 1 to 1.5 mg/kg IV) should be administered. (See "[Anesthesia for the patient with preeclampsia](#)", [section on 'Intraoperative magnesium'](#).)

The use of magnesium to reduce dose of NMBA can result in unpredictable neuromuscular blockade and increase the incidence of residual paralysis, which can increase postoperative pulmonary complications. In a meta-analysis of 21 randomized trials, the median dose reduction of NMBA after administration of magnesium was approximately 25 percent (interquartile range 14.7 to 31 percent), without an increase in complications [14]. Here was wide variation in the dose of magnesium and the overall quality of evidence was moderate, with high statistical heterogeneity.

- Hypercalcemia causes a reduced response to the administration of nondepolarizing NMBAs as calcium triggers the release of acetylcholine into the NMJ and enhances the excitation-contraction coupling of the myocytes [15]. Thus, hypercalcemic patients may require larger doses of nondepolarizing NMBA to achieve the desired level of neuromuscular blockade.
- Hypocalcemia should be confirmed by measuring ionized calcium, and excluding falsely low levels of calcium due to low plasma albumin levels. Plasma calcium can have antagonistic actions at the neuromuscular junction; at the presynaptic membrane, it can decrease the amplitude of depolarization, thus antagonizing nondepolarizing block [16]. At the postsynaptic membrane, calcium decreases the degree of depolarization produced by acetylcholine, thus potentiating nondepolarizing block [17]. For these reasons, the overall effect of calcium on neuromuscular transmission is unpredictable [15].
- Hypokalemia may prolong the duration of nondepolarizing agents, but has no effect on the duration of depolarizing block [18].

- **Acidosis** – Acidosis, whether metabolic or respiratory in origin, can prolong the effects of NMBAs, by increasing NMBA affinity for postjunctional nAChRs. Acidosis also causes a relative intracellular hypocalcemia, which can prolong neuromuscular block [16,19]. Conversely, alkalosis can shorten the duration of neuromuscular blockade produced by nondepolarizing muscle relaxants, but will not affect the duration of action of depolarizing muscle relaxants [18]. These perturbations can occur once pH becomes lower than 7.3 or greater than 7.51 [19].
- Drug interactions** — A number of drugs can either enhance or inhibit the response to NMBAs, and may require modification of the dose or interval of administration, guided by monitoring.
- **Combinations of NMBAs** – In some clinical circumstances, [succinylcholine](#) may be administered before or after a nondepolarizing NMBA, or two different nondepolarizing NMBAs may be administered in sequence. Combinations of NMBAs can affect the resulting degree of neuromuscular block, and subsequent management should be guided by the use of a neuromuscular function monitor.
- Prior administration of any nondepolarizing neuromuscular blocking drug has a substantial antagonistic effect on the subsequent depolarizing block induced by [succinylcholine](#). As an example, if a defasciculating dose of a nondepolarizing NMBA is administered prior to succinylcholine, the dose of succinylcholine must be increased [20].
- In contrast, the effect of prior administration of [succinylcholine](#) on subsequent administration of a nondepolarizing neuromuscular block depends on the drug used. Studies have reported that administration of succinylcholine prior to a nondepolarizing NMBA does not affect the potency of mivacurium [21] or [rocuronium](#) [22], but speeds the onset, increases the potency, and prolongs the duration of action of [vecuronium](#) and [cisatracurium](#) [23].
- Combining two nondepolarizing NMBAs of the same structural class, such as [rocuronium](#) and [vecuronium](#), produces an additive interaction [24]. Combining two nondepolarizing NMBAs from different structural classes, such as rocuronium and [cisatracurium](#), results in a synergistic response [25].
- **Inhaled anesthetics** – Inhaled anesthetics inhibit nAChRs and potentiate neuromuscular blockade with nondepolarizing NMBAs. This potentiation depends on the type of volatile anesthetic ([desflurane](#) > [sevoflurane](#) > [isoflurane](#) > [nitrous oxide](#)), the concentration, and the duration of exposure [26-28].
- **Antibiotics** – Tetracyclines, aminoglycosides, polymyxins, and [clindamycin](#) potentiate neuromuscular blockade through inhibition of acetylcholine (ACh) release or desensitization of postjunctional nAChRs to ACh [29]. In practice, this interaction is most relevant during maintenance of anesthesia. Antibiotics typically are given after a dose of NMBA has already been administered for induction of anesthesia. Thus the interaction between such antibiotics and nondepolarizing NMBAs must be considered when redosing NMBAs. Appropriate (objective) monitoring is particularly helpful in this setting.
- **Antiseizure drugs** – Patients receiving chronic treatment are relatively resistant to nondepolarizing NMBAs due to accelerated clearance of these drugs [30,31].
- **Lithium** – Patients who take [lithium](#) can have a prolonged response to both depolarizing and nondepolarizing NMBAs [32]. Lithium structurally resembles other cations, such as sodium, potassium, magnesium, and calcium. This resemblance to such cations activates potassium channels, inhibiting neuromuscular transmission.
- **Antidepressants** – In laboratory studies, [sertraline](#) and [amitriptyline](#) inhibit butyrylcholinesterase [33], and prolonged paralysis after administration of mivacurium has been reported in a patient who was chronically taking sertraline [34].
- **Local anesthetics** – Local anesthetics (LAs) may enhance the effects of both depolarizing and nondepolarizing NMBAs through pre- and post-synaptic interactions at the neuromuscular junction (NMJ) [35,36]. Administration of LAs for regional anesthesia may result in blood levels high enough to potentiate NMBA-induced neuromuscular block. Epidurally administered levobupivacaine [37] and [mepivacaine](#) [38] potentiate amino-steroidal NMBAs and delay recovery from neuromuscular blockade.

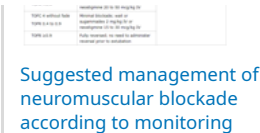
Data on the effects of [lidocaine](#) on neuromuscular block are conflicting. Whereas one study reported that epidural lidocaine potentiated vecuronium-induced neuromuscular block [39], several studies have reported no effect of intravenous (IV) lidocaine on the onset, duration, or recovery from rocuronium-induced neuromuscular block [40-42].

CLINICAL USE OF NEUROMUSCULAR BLOCKING AGENTS

Endotracheal intubation — The most common indication for the use of NMBAs in anesthesia is to facilitate endotracheal intubation. The goal of muscle paralysis for endotracheal intubation is to achieve ideal intubation conditions, which are dependent on the depth of anesthesia, airway anatomy, and the experience of the anesthesiologist, in addition to the depth of neuromuscular block ([table 6](#)). More intense block is associated with better intubation conditions; a dose of 2 X ED95 for a given NMBA should be sufficient for routine intubation [43,44].

Suggested management of neuromuscular blockade according to monitoring		
Monitoring method	Interpretation/management	Comments
Table 1	Respiratory conditions stable	Continue current NMBA dose
Table 2	Respiratory conditions stable	Continue current NMBA dose
Table 3	Respiratory conditions stable	Continue current NMBA dose
Table 4	Respiratory conditions stable	Continue current NMBA dose
Table 5	Respiratory conditions stable	Continue current NMBA dose
Table 6	Respiratory conditions stable	Continue current NMBA dose
Table 7	Respiratory conditions stable	Continue current NMBA dose
Table 8	Respiratory conditions stable	Continue current NMBA dose
Table 9	Respiratory conditions stable	Continue current NMBA dose
Table 10	Respiratory conditions stable	Continue current NMBA dose
Table 11	Respiratory conditions stable	Continue current NMBA dose
Table 12	Respiratory conditions stable	Continue current NMBA dose
Table 13	Respiratory conditions stable	Continue current NMBA dose
Table 14	Respiratory conditions stable	Continue current NMBA dose
Table 15	Respiratory conditions stable	Continue current NMBA dose
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Table 98	Respiratory conditions stable	Continue current NMBA dose
Table 99	Respiratory conditions stable	Continue current NMBA dose
Table 100	Respiratory conditions stable	Continue current NMBA dose

The use of NMBAs may reduce the incidence of postintubation hoarseness and airway injury [5,45]. A 2018 meta-analysis of randomized controlled trials that compared endotracheal intubation with NMBAs and without NMBAs reported increased risks of difficult intubation and an increased risk of upper airway discomfort and injury if NMBAs were avoided for intubation, but the overall quality of the evidence was moderate to low [46].



Suggested management of neuromuscular blockade according to monitoring

Table 6 - larger image below

Succinylcholine is commonly used for endotracheal intubation because of its reliable and rapid onset of neuromuscular blockade. Skeletal muscle relaxation can be assumed immediately after resolution of the fasciculations that occur with succinylcholine (see '[Adverse effects of succinylcholine](#)' below). After administration of succinylcholine, a nondepolarizing NMBA should not be administered until return of neuromuscular function has been documented, as confirmation of normal pseudocholinesterase activity. (See '[Butyrylcholinesterase \(pseudocholinesterase\) deficiency](#)' below.)

Nondepolarizing NMBAs do not cause fasciculations. Therefore, we suggest the use of an objective neuromuscular function monitor to determine when optimal neuromuscular blockade is achieved after administration of nondepolarizing NMBAs for endotracheal intubation. Central muscle groups (eg, the diaphragm and laryngeal adductors) are blocked faster than peripheral muscle groups (eg, adductor pollicis) after the administration of NMBA [47]. Therefore, if optimal neuromuscular blockade (defined as total suppression of evoked responses to train-of-four [TOF] stimulation) is confirmed by monitoring adductor pollicis stimulation, pharyngeal muscles are usually blocked as well, and optimal intubating conditions are achieved.

The onset time of nondepolarizing NMBAs can be shortened by increasing the dose. As an example, **rocuronium** can be used for rapid sequence induction and intubation within 60 seconds by administration of twice the usual intubation dose (or four times the ED₉₅). (See '[Rocuronium](#)' below and '[Rapid sequence induction and intubation \(RSII\) for anesthesia](#)', section on '[Nondepolarizing NMBAs](#)'.)

Rapid onset of paralysis can also be achieved by administration of a priming dose (10 percent of the intubating dose) several minutes prior to the intubating dose [48]. Since the priming dose of NMBA may cause signs of paralysis, including diplopia, blurry vision, difficulty swallowing, and risk of aspiration, we do not routinely administer a priming dose of NMBA [49].

Facilitation of surgery — Nondepolarizing NMBAs may be used to provide muscle relaxation that improves surgical conditions. However, many operations can be performed without the need to use any NMBAs, and administration of NMBAs should be individualized [50,51]. Neuromuscular blockade may improve surgical conditions during laparoscopic, robotic, abdominal, and thoracic procedures, by reducing patient movement, muscle tone, and breathing or coughing against the ventilator [2,52-55]. NMBAs also allow lower insufflation pressures during laparoscopy, but the relationship between insufflation pressure and surgical exposure is unclear. The benefits of deep versus moderate muscle relaxation during laparoscopy are also unclear [56-59]. Many of the theoretical benefits of neuromuscular blockade can be achieved by making sure that patients are adequately anesthetized at all times during surgery ([table 6](#)).

- **Moderate neuromuscular block** – When neuromuscular blockade is indicated, for most surgical procedures we aim for moderate levels of blockade (ie, one to three twitches using an objective peripheral nerve stimulator ([figure 1](#))). (See '[Monitoring neuromuscular blockade](#)', section on '[Train-of-four](#)'.)

Once neuromuscular block begins to recover from the deep level of paralysis required for endotracheal intubation, a moderate level of block may be maintained with interval dosing or an infusion of nondepolarizing NMBA, guided by monitoring. When guided by objective monitoring, administration of NMBAs by infusion may reduce the total dose required for surgical relaxation [60].

At the conclusion of abdominal surgery, a moderate level of relaxation of abdominal muscles is often sufficient to facilitate closure of the wound. A deep level of block should be avoided at this stage of the operation to allow rapid and complete reversal of the NMBA. If necessary, the anesthetic can be deepened temporarily (ie, with inhalation agent or **propofol**) to facilitate closure without administration of NMBA.

- **Deep neuromuscular blockade** – Deep levels of neuromuscular blockade (zero twitches and post tetanic count >1) may be warranted in specific settings, in addition to ensuring adequate depth of anesthesia. As an example, complete paralysis may be indicated during delicate dissection under the microscope during craniotomy, when any unexpected movement must be avoided. (See '[Anesthesia for intracranial neurovascular procedures in adults](#)', section on '[Patient immobility](#)'.)

AVOIDANCE OF RESIDUAL NEUROMUSCULAR BLOCKADE

One primary goal for anesthetic management is to ensure complete recovery of neuromuscular function at the end of anesthesia (ie, train-of-four ratio [TOFR] ≥0.9).

Consequences of residual block — Residual neuromuscular block after administration of NMBAs is an important risk factor for anesthesia-related morbidity and mortality. Even minor degrees of residual block are associated with weakness of upper airway muscles, airway obstruction, increased risk of aspiration, and unpleasant muscle weakness. Incomplete postoperative neuromuscular recovery can also cause prolonged recovery room stay, hypoxemia and airway obstruction, awareness during emergence from anesthesia, and increased postoperative pulmonary complications [61-63].

A prospective international observational study including approximately 22,000 patients who underwent general anesthesia for noncardiac surgery at 1 of over 200 hospitals reported that the use of NMBAs was independently associated with an increase in postoperative pulmonary complications within 28 days of surgery [64]. In contrast with other literature, in this study neither the use of neuromuscular monitoring nor administration of reversal agents was associated with a reduction in postoperative pulmonary complications. The utility of this study has been questioned and a number of concerns have been raised, including poorly described reversal administration and the utilization of uncalibrated acceleromyographic data to confirm recovery [65-68]. A post hoc analysis of the study found that raising the threshold for extubation from a TOFR ≥ 0.9 to TOFR ≥ 0.95 was associated with a lower incidence of postoperative pulmonary complications [69]. These findings reinforce the need to calibrate accelerometers prior to administration of NMBAs, and support the need to increase the reversal threshold for extubation if calibration is not performed. (See ["Monitoring neuromuscular blockade"](#), section on ["Acceleromyography \(AMG\)"](#).)

Incidence of residual block — Multiple studies have reported that residual neuromuscular block is a common problem [70-72]. In a meta-analysis of 24 randomized and observational studies including 3375 patients anesthetized between 1979 and 2005, residual block (TOFR <0.9) occurred in 41 percent of patients who received intermediate acting NMBAs [73]. A 2020 meta-analysis of 53 observational and randomized trials found that for patients who received intermediate-acting NMBAs (ie, [atracurium](#), [cisatracurium](#), [mivacurium](#), [vecuronium](#), or [rocuronium](#)), the use of quantitative neuromuscular monitoring was associated with a reduced incidence of postoperative residual neuromuscular block, compared with qualitative (subjective) monitoring, or no monitoring at all (11.5 versus 30.6 versus 33.1 percent, respectively) [74]. However, the quality of evidence was judged to be very low.

Strategy for avoiding residual block — A suggested strategy for avoidance of residual neuromuscular blockade is as follows:

- Use intermediate or short acting NMBAs, rather than long acting NMBAs (eg, pancuronium) whenever possible
- Avoid deep neuromuscular blockade (ie, train-of-four count [TOFC] = 0, post tetanic count >1) when clinically appropriate
- Use objective neuromuscular monitoring whenever possible to guide administration of NMBAs and assess recovery of neuromuscular function for all patients who receive NMBAs
 - If only a peripheral nerve stimulator is available, ensure electrode placement is not causing direct muscle stimulation ([figure 2](#) and [figure 3](#)).
- Avoid monitoring facial muscles as this will overestimate the degree of recovery. If the arms are not accessible due to surgical positioning, monitor the facial muscles but transition to the arms once they become available. (See ["Monitoring neuromuscular blockade"](#), section on ["Nerves that may be monitored"](#).)
- Administer reversal agents in appropriate doses, and guided by the degree of recovery from neuromuscular block ([table 6](#)). (See ["Reversal of neuromuscular block"](#) below.)
 - Administer no reversal agents if adequate spontaneous recovery has been achieved, as demonstrated by a quantitative TOFR >0.9.
 - Administer [neostigmine](#), in appropriate doses, only if spontaneous recovery has reached a TOFC = 4.
- If spontaneous recovery has not reached a TOFC = 4 (as assessed at the adductor pollicis muscle), use [sugammadex](#) rather than [neostigmine](#) for reversal of steroidal NMBAs. If sugammadex is unavailable (or if using benzylisoquinolinium NMBAs), wait for spontaneous recovery to achieve a TOFC = 4 before administering neostigmine.
- Extubate the trachea only after a TOFR ≥ 0.9 is achieved (when objective monitors are available).
- If a quantitative monitor is not available, administer [neostigmine](#) only when TOFC = 4 (as assessed at the adductor pollicis muscle) and allow adequate time for full reversal of neuromuscular blockade, at least 10 minutes after neostigmine administration, before tracheal extubation [75,76]. (See ["Anticholinesterases"](#) below.)

As the operation nears completion, allow the patient to recover to a TOFC of 4 by the time the surgeon starts to close the wound. When [neostigmine](#) is administered at this time, one would expect to be ready to extubate the trachea in 10 minutes or so, once the drapes are coming down. If the patient moves or became dyssynchronous with the ventilator near the end of the operation, administer [propofol](#) rather than NMBA. This timing requires familiarity with the surgeon and each individual practice.

These recommendations are consistent with 2023 guidelines from the American Society of Anesthesiologists ([table 7](#)) [77].

REVERSAL OF NEUROMUSCULAR BLOCK

A train-of-four ratio (TOFR) ≥ 0.9 should be achieved prior to tracheal extubation following administration of NMBAs [78]. Reversal of neuromuscular block after administration of NMBAs can occur by spontaneous recovery, or by administration of reversal agents. Reversal agents include

Recommendations from the ASA guidelines for neuromuscular blockade	
1.	When neuromuscular blocking drugs are administered, we recommend against clinical assessment alone to avoid residual neuromuscular blockade, due to the unreliability of the assessment.
2.	We recommend quantitative monitoring over qualitative assessment to avoid residual neuromuscular blockade.
3.	When using quantitative monitoring, we recommend confirming a train-of-four ratio greater than or equal to 0.9 before extubation.
4.	We recommend using the adductor pollicis muscle for neuromuscular monitoring.
5.	We recommend against using non-muscles for neuromuscular monitoring.
6.	We recommend appropriate use of monitoring at sites, conditions, and stimulus depths of administration that are unlikely to result in incomplete or unresponsive, to avoid residual neuromuscular blockade. ^a
7.	We suggest neostigmine as a reasonable alternative to sugammadex at minimal depth of neuromuscular blockade.
8.	To avoid residual neuromuscular blockade when atracurium or cisatracurium are administered and qualitative assessment is used, we suggest enflurane with neostigmine or rocuronium with sugammadex. In the absence of quantitative monitoring, at least 10 minutes should elapse from emergence to extubation. When quantitative monitoring is utilized, extubation can be done as soon as a train-of-four ratio greater than or equal to 0.9 is confirmed before extubation.
^a Deep sedation must be greater than or equal to 3 and train-of-four count to extubation: train-of-four count 0 to 3: extubate; train-of-four count 4 and greater: train-of-four ratio less than 0.4: extubate; train-of-four ratio 0.4 to less than 0.9: extubate and administer first intubation dose again. (See table 6 , table 7 , and table 8 .)	
^b In 2023, American Society of Anesthesiologists (ASA) guidelines for monitoring and management of neuromuscular blockade, a report to the American Society of Anesthesiologists and British Society of Anaesthetists, is available at https://www.asahq.org/ASA/education/continuing-education/neuromuscular-blockade-2023 .	
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anticholinesterases and [sugammadex](#), the only available selective relaxant reversal agent, which is specific for aminosteroid nondepolarizing NMBAs.

Recommendations from the ASA guidelines for neuromuscular blockade

The recommended doses of the commonly used reversal agents and the time course of recovery from various levels of rocuronium-induced neuromuscular block are shown in a table ([table 6](#)).

Table 7 - larger image below

Reversal agents can rarely cause adverse effects that require treatment (bradycardia, bronchospasm, and very rarely anaphylaxis or cardiac arrest). Thus we strongly recommend physiologic monitoring and vigilance throughout emergence from anesthesia whenever reversal agents are used. (See '[Sugammadex versus neostigmine](#)' below.)

Spontaneous recovery — Reversal agents should only be omitted when clinicians can confirm adequate recovery from neuromuscular blockade to a TOFR of 0.9 or greater using quantitative monitoring. The response to single intubating doses of NMBAs is variable among patients and responses can be difficult to predict. As an example, a large trial found that after a single intubating dose of an intermediate-acting NMBA, 37 percent of patients had not spontaneously recovered to a TOFR of 0.9 two hours after administration [79].

Anticholinesterases — Anticholinesterase agents work by increasing the availability of acetylcholine at the neuromuscular junction, to compete with nondepolarizing NMBAs for nicotinic acetylcholine receptors (nAChRs) and restore neuromuscular transmission. The anticholinesterases available for reversal of NMBA are [neostigmine](#) and edrophonium. Neostigmine is usually preferred, despite a slower onset of action than edrophonium, because neostigmine has higher affinity for acetylcholine receptors. Both drugs have a duration of action of 60 to 120 minutes [80].

Blockade of cholinesterase occurs at all cholinergic sites, including muscarinic receptors, such that these drugs have parasympathomimetic effects. Therefore, anticholinergic, antimuscarinic drugs (ie, [atropine](#) or [glycopyrrolate](#)) are administered along with anticholinesterases to prevent bradycardia and gastrointestinal side effects.

- **Neostigmine** – [Neostigmine](#) is the anticholinesterase agent most commonly used for reversal of neuromuscular blockade. [Glycopyrrolate](#) is routinely administered along with neostigmine (0.2 mg glycopyrrolate IV for each 1 mg neostigmine IV) as they have a similar onset of action.

The rate of reversal of neuromuscular block with [neostigmine](#) is variable. Factors that affect the speed of reversal include the depth of existing block, the dose of neostigmine, and the other factors that can affect the degree of neuromuscular block (eg, inhalation anesthesia, hypothermia, acidosis, electrolyte disturbances, hypercarbia, patient age) [81,82]. (See '[Factors that affect the response to neuromuscular blocking agents](#)' above.)

Reversal of neuromuscular block with [neostigmine](#) takes longer during inhalation anesthesia than during intravenous (IV) anesthesia with [propofol](#). In one study, 160 patients were randomly assigned to maintenance of anesthesia with [sevoflurane](#) or propofol, and reversal of [rocuronium](#) with neostigmine at different levels of block. The median time from administration of neostigmine at a train-of-four count (TOFC) of 3 to recovery to a TOFR of 0.9 was 15.6 minutes (7.3 to 43.9) during sevoflurane anesthesia and was 5.4 minutes (1.6 to 8.6) during propofol anesthesia [82]. The median time to recovery if neostigmine was administered at TOFC of 4 was 9.7 minutes (5.1 to 26.4) during sevoflurane anesthesia, and 4.7 minutes (1.3 to 7.2) during propofol anesthesia.

If sufficient spontaneous recovery (train-of-four [TOF] >0.9) can be confirmed through quantitative monitoring, [neostigmine](#) is unnecessary. If quantitative monitoring is not available or this level of recovery has not been achieved, neostigmine (or [sugammadex](#)) should be administered in an appropriate dose based on the degree of neuromuscular block at the time of administration ([table 6](#)).

- [Neostigmine](#) is ineffective at reversing deep levels of neuromuscular blockade (TOFC = 0, post-tetanic count [PTC] ≥ 1 , TOFC = 0) [81].
- [Neostigmine](#) should be administered at a TOFC of 4 (shallow levels of blockade) [83], and the patient should be extubated no sooner than 10 minutes after administration of neostigmine. At a TOFC of 4, neostigmine doses of 20 to 50 mcg/kg may be sufficient to effect adequate recovery [84]. Higher doses than these are no more effective, and increase the risk of a direct neuromuscular blocking effect of the drug. A reduced dose of 15 to 30 mcg/kg may be used if there is minimal block using an objective neuromuscular monitor (ie, TOFR >0.4 and <0.9) or if there is a TOFC of 4 without fade using a peripheral nerve stimulator ([table 8](#)) [77].
- If the patient has a TOFC of 1 to 3 at the conclusion of an operation, wait for spontaneous recovery to TOFC of 4 prior to administration of [neostigmine](#).
- If reversal is inadequate after the initial dose of [neostigmine](#), additional neostigmine may be administered up to a total dose of 70 mcg/kg (maximum 5 mg) IV. If [rocuronium](#) or [vecuronium](#) was used, it is reasonable to administer [sugammadex](#) 1 or 2 mg/kg IV in this setting, preferably guided by quantitative neuromuscular monitoring. (See '[Sugammadex](#)' below.)
- The use of [neostigmine](#) in the absence of NMBAs can result in paradoxical neuromuscular weakness [85,86]. Administration of neostigmine during recovery from neuromuscular blockade, even at TOFR >0.9, is unlikely to result in clinically significant weakness. It appears that nAChR occupancy by the remaining nondepolarizing molecules provides a protective effect from the depolarizing actions of excess acetylcholine. This was demonstrated by a study in which 90 anesthetized patients who had recovered to a TOFR >0.9 after a single dose of [rocuronium](#) were randomly assigned to receive neostigmine 40 mcg/kg IV or [saline](#) [87]. There were no differences in

TOFR five minutes after reversal or on admission to the postanesthesia care unit, and no differences in postoperative weakness, hypoxemia, or airway obstruction. Neostigmine-induced decrements in neuromuscular function should almost **never** occur clinically [86].

- **Edrophonium** – Edrophonium has a faster onset (one to two minutes) than [neostigmine](#) and is usually coadministered with [atropine](#) (7 to 14 mcg/kg IV), which has a similar onset of action, to block antimuscarinic effects. Binding of edrophonium to acetylcholinesterases is much weaker than the neostigmine, and it should only be used at shallow levels of neuromuscular blockade. The standard dose of 0.5 to 1 mg/kg results in a similar duration of antagonism as neostigmine [80].

Sugammadex — [Sugammadex](#) is a gamma-cyclodextrin that encapsulates and subsequently inactivates steroidal NMBAs. Affinity is greatest for [rocuronium](#), followed in decreasing order of affinity by [vecuronium](#), and pancuronium [88]. Sugammadex has no effect on [succinylcholine](#) or the benzylisoquinolinium NMBAs.

After administration, binding occurs in plasma, leading to a rapid decrease in the plasma levels of unbound NMBA molecules. This results in a concentration gradient between the free NMBA molecules at the neuromuscular junction (NMJ) and in the plasma, leading to a net transfer of unbound NMBA back into the plasma, where it is encapsulated by unbound [sugammadex](#). This rapid decrease in free NMBA at the NMJ results in rapid and effective antagonism and restoration of normal transmission and neuromuscular function. Sugammadex reversal produces a more complete and faster recovery than [neostigmine](#) or edrophonium reversal [89-91].

- **Dosing** – The dose of [sugammadex](#) should be based on the level of neuromuscular block.
 - At a deep level of neuromuscular blockade (TOFC = 0, PTC 1 to 2), a dose of 4 mg/kg results in recovery to a TOFR ≥ 0.9 within five minutes [92].
 - At a TOFC 2 to 4 a dose of [sugammadex](#) of 2 mg/kg IV results in a TOFR ≥ 0.9 in three to four minutes [93].
 - In emergency situations in which intubation and ventilation are unexpectedly difficult or impossible, 16 mg/kg [sugammadex](#) can reverse profound rocuronium-induced blockade within three minutes [94]. Whereas rapid reversal of neuromuscular block may be helpful in this setting, a number of other factors may affect the restoration of spontaneous ventilation (eg, coadministered anesthetics and opioids, patient factors). (See "[Management of the anatomically difficult airway for general anesthesia in adults](#)", section on 'The failed airway'.)
 - If reversal of aminosteroid induced block (ie, after [rocuronium](#) or [vecuronium](#)) is inadequate after administration of up to 70 mcg/kg or 5 mg IV of [neostigmine](#), it is reasonable to administer 1 to 2 mg/kg [sugammadex](#) IV, guided by quantitative neuromuscular monitoring. Optimal dosing of sugammadex in this setting has not been determined.
 - Importantly, the response to the doses of [sugammadex](#) based on the TOFC may vary. Quantitative monitoring should be used to confirm adequate reversal prior to extubation, even when reversing with sugammadex, as residual weakness is still possible. This was demonstrated in a single-center prospective dose-finding study in 97 patients who underwent cardiac surgery and received sugammadex for reversal of rocuronium-induced neuromuscular block [95]. After receiving the manufacturer recommended doses (which are described above), 13 percent of patients had residual weakness (ie, TOFR < 0.9), and two patients had recurrent paralysis.

Administration of excessive doses of [sugammadex](#) should be avoided, because the incidence of anaphylaxis with sugammadex is proportional to the dose administered [96], excessive administration may prevent or delay subsequent reestablishment of neuromuscular block with steroidal NMBAs, and sugammadex is expensive.

- **Patients with kidney dysfunction** – The sugammadex-NMBA complex is excreted in the urine unchanged [97]. No dosing adjustments are required for patients with creatinine clearance (CrCl) ≥ 30 mL/minute. [Sugammadex](#) is not approved by the US Food and Drug Administration (FDA) for patients with severe kidney dysfunction (CrCl < 30 mL/minute), primarily due to concerns about the possibility of recurrence of paralysis if the complex dissociates. There is also concern that prolonged exposure to sugammadex could increase the risk of anaphylaxis. We antagonize amino steroidal-induced block with sugammadex in patients with eGFR < 30 mL/minute under the guidance of objective neuromuscular monitoring. However, use of sugammadex in patients with severe kidney dysfunction is an area of active research involving a vulnerable patient population, and continued vigilance in the postoperative period is certainly warranted.

[Sugammadex](#) is widely used, including in patients with kidney dysfunction. A study of registry data from 2016 to 2022 for approximately 250,000 cases from the Multicenter Perioperative Outcomes Group compared reversal strategies with [rocuronium](#)/sugammadex, rocuronium/[neostigmine](#), and [cisatracurium](#)/[neostigmine](#) [98]. Rocuronium/sugammadex was used in over 95 percent of patients overall, and in over 50 percent of patients with eGFR < 30 mL/minute. However, the literature on the safety of reversal with sugammadex in patients with kidney dysfunction is conflicting, and further study is required.

- In a single institution randomized trial including 49 patients with CrCl < 30 mL/minute who received neuromuscular block with [cisatracurium](#) followed by reversal with [neostigmine](#) versus [rocuronium](#) reversed with [sugammadex](#), reversal was faster in patients who received rocuronium/sugammadex, and there were no major adverse events in either group [99].
- Similarly, in two multicenter retrospective reviews of [sugammadex](#) reversal in surgical patients with end-stage kidney disease (total 355 patients) there were no cases of recurrence of neuromuscular blockade, and no cases of anaphylaxis [100,101].

- In contrast, in a propensity matched cohort study of patients with chronic kidney disease from the TriNetX international research database for 2004 to 2024, neuromuscular blockade and reversal with [rocuronium/sugammadex](#) was associated with increased risk of cardiopulmonary complications (ie, respiratory failure, hypertensive crisis, heart failure, and 30 day mortality), compared with [cisatracurium/neostigmine](#) [102]. Conclusions are limited by the retrospective nature of the study, lack of data on the types of surgery performed, lack of details on anesthetic and perioperative management, and inability to determine whether results relate to the choice of NMBA or the reversal agent.
- **Drug interactions** – [Sugammadex](#) can bind and inhibit oral contraceptives. Patients should use alternative contraceptive techniques for seven days after exposure to sugammadex [103]. Reestablishing neuromuscular blockade with steroidal NMBA within three hours of sugammadex administration requires increased doses [104].
- **Adverse effects** – [Sugammadex](#) has been associated with several types of adverse events.

- **Hypersensitivity reactions** – [Sugammadex](#) has increasingly been implicated as a cause of perioperative anaphylaxis. The incidence of anaphylaxis after exposure to sugammadex is unclear, and has been reported to occur in 1:3500 to 1:64,000 exposures. (See "[Perioperative anaphylaxis: Allergy evaluation and prevention of recurrent reactions](#)", section on 'Sugammadex'.)

Drug hypersensitivity reactions are the result of immune or inflammatory cell stimulation by the drug, and may present during anesthesia with flushing; rash; urticaria; angioedema of the face, extremities, or larynx; wheezing; and/or hypotension. Anaphylaxis is the most severe form of immediate hypersensitivity reaction. The diagnosis is clinical and based on rapid onset and progression of signs and symptoms that typically include hypotension, bronchospasm, laryngeal edema, and tachycardia. Classification and clinical manifestations of drug hypersensitivity are discussed in detail separately. (See "[Drug hypersensitivity: Classification and clinical features](#)" and "[Perioperative anaphylaxis: Recognition and emergency management](#)", section on 'Recognition of anaphylaxis'.)

Hypersensitivity reactions to [sugammadex](#) typically appear rapidly after IV administration (ie, usually within a minute but sometimes up to several minutes later) and can occur with first-time administration. Risk factors have not been identified, though the incidence of some types of reactions appears to be dose dependent [105]. Reactions can involve clinically significant airway edema and bronchospasm, and if this develops after tracheal extubation, reintubation may be required. Hypersensitivity reactions and anaphylaxis to sugammadex are discussed in several other topics. (See "[Perioperative anaphylaxis: Recognition and emergency management](#)", section on 'Recognition of anaphylaxis' and "[Perioperative anaphylaxis: Allergy evaluation and prevention of recurrent reactions](#)", section on 'Sugammadex'.)

- **Cardiac arrhythmias** – Cardiac arrhythmias, including marked bradycardia and asystole, may occur after administration of [sugammadex](#) [106]. Thus, full electrocardiogram (ECG) monitoring should be continued during and after administration of sugammadex, and [atropine](#) and other vasoactive drugs should be immediately available [107]. From 2009 to 2017, 138 cases of serious cardiac adverse events with nine deaths after administration of sugammadex were reported to the US Food and Drug Administration adverse events reporting system, including bradycardia (n = 66), cardiac arrest (n = 39), ventricular fibrillation (n = 10), and ventricular tachycardia (n = 8) [108]. The reported incidence of sugammadex-induced bradycardia is approximately 1 percent [109], and the etiology is unknown. While sugammadex-induced bradycardia is typically transient, it has precipitated cardiac arrest and hemodynamic collapse in some cases [106,109,110].
- **Others** – Small increases in coagulation parameters have been reported after [sugammadex](#) administration, but clinically significant bleeding has not been reported [111]. There have been several reports of laryngospasm that occurred near the time of sugammadex administration [112-114]. While a direct causal relationship has not been established, vigilance is warranted in maintaining airway patency as neuromuscular function is restored.

Sugammadex versus neostigmine — For patients who receive steroidal NMBAs (eg, [rocuronium](#) or [vecuronium](#)) and require reversal, we suggest using [sugammadex](#) rather than [neostigmine](#). However, sugammadex is significantly more expensive than neostigmine. Where cost is an issue, it is reasonable to restrict the use of sugammadex to patients who are at higher risk of complications of postoperative residual neuromuscular block (eg, patients >80 years of age or who undergo thoracic surgery), and/or patients who need reversal from TOFC ≤3.

Benefits of [sugammadex](#) compared with [neostigmine](#) include the following:

- [Sugammadex](#) reverses neuromuscular blockade more quickly and reliably than [neostigmine](#), and can be used to reverse deeper levels of block [115,116].
- [Sugammadex](#) reversal is associated with reduced incidence of residual neuromuscular blockade [115,117].
- [Sugammadex](#) reversal probably reduces postoperative pulmonary complications, particularly in high risk patients.
- In a single institution randomized trial of 180 high risk older adults (ASA physical status 3 or 4, ≥75 years of age) who underwent nonemergency surgery with [rocuronium](#) induced neuromuscular blockade, there was a trend towards improved postoperative pulmonary outcomes in patients who received [sugammadex](#), though the results did not reach statistical significance [118]. The rate of radiograph confirmed pneumonia within seven days was lower in patients who received sugammadex (2.4 versus 9.6 percent), though

the number of events was small. Conclusions are limited by the timing of reversal; the agents were administered at a deeper level of block than we recommend when using [neostigmine](#) (four twitches with TOF monitoring, TOFR ≥ 0.4). (See '[Anticholinesterases](#)' above.)

- In a multicenter matched cohort study of over 45,000 surgical patients, compared with [neostigmine](#), reversal with [sugammadex](#) was associated with a lower incidence of pneumonia (1.3 versus 2.2 percent, odds ratio [OR] 0.7 [95% CI 0.44-0.62]) and reduced risk of respiratory failure (0.8 versus 1.7 percent, OR 0.45 [95% CI 0.37-0.56]) [119]. Median patient age was 58 years; 55 percent were ASA physical status 3.
- In contrast, in a single institution cohort study that compared the incidence of postoperative pulmonary complications (PPCs) before and after an institutional change from using [neostigmine](#) to using [sugammadex](#) for routine reversal of NMBAs, the incidence of PPCs was similar before and after the switch [120]. Conclusions from this study are limited by the possibility of other practice changes over time, and inconsistency or absence of data on neuromuscular monitoring.
- [Sugammadex](#) probably does not result in faster gastric emptying compared with [neostigmine](#). In a trial that included 60 adults undergoing colorectal surgery, gastric emptying time was similar in patients receiving sugammadex versus neostigmine [121]. However, the time to first bowel movement was numerically shorter in patients receiving sugammadex (44.3 versus 61.0 hours, with a difference of 16.7 hours [95% CI 2.3-31.1 hours]).

Serious intraoperative adverse effects and complications of [sugammadex](#) and [neostigmine](#) are rare, and both drugs appear to be similarly safe. This is supported by a single institution retrospective review of clinically important adverse events in approximately 90,000 patients who underwent general, cardiac, or pediatric surgery and who received sugammadex or neostigmine for reversal of neuromuscular blockade at the end of surgery [122]. The incidence of a composite outcome of adverse events requiring pharmacologic intervention (ie, bradycardia, anaphylaxis, bronchospasm, and cardiac arrest) was slightly higher with sugammadex (3.4 percent) than neostigmine (3.0 percent), based primarily on differences in bronchospasm and bradycardia, which are usually easily treated. Anaphylaxis and cardiac arrest were extremely rare with both drugs (0.004 to 0.02 percent). The difference in the composite outcome resulted in a number needed to harm of 250 patients, meaning 250 patients would need to be given neostigmine rather than sugammadex to avoid one of these events.

In a meta-analysis of 10 randomized trials (959 patients) that compared NMBA reversal with [sugammadex](#) versus [neostigmine](#), there was no significant difference in the incidence of one or more serious adverse event or a composite of serious adverse events [90,123,124]. However, in a large retrospective database study, use of sugammadex rather than neostigmine with [glycopyrrolate](#) was associated with reduced incidence of postoperative urinary retention [124].

CLASSIFICATION OF NEUROMUSCULAR BLOCKING AGENTS

NMBAs are classified according to their interaction with nicotinic acetylcholine receptors (nAChRs) at the neuromuscular junction (NMJ). [Succinylcholine](#), the only depolarizing NMBA in use, produces skeletal muscle relaxation by binding directly with nAChRs to cause prolonged depolarization. Nondepolarizing NMBAs act as competitive antagonists (competing with acetylcholine [ACh] for the binding sites at the nAChRs), preventing the initiation of action potential. The type of neuromuscular block determines the response to peripheral nerve stimulation used for monitoring the effects of NMBAs. (See "[Monitoring neuromuscular blockade](#)".)

The NMJ consists of the presynaptic motor neuron, a synaptic cleft or gap, and the postsynaptic surface of the myocyte ([figure 4](#)) [15]. As an electrical signal in a motor nerve reaches the presynaptic nerve terminal, depolarization occurs, and acetylcholine (ACh) is released via calcium channel-mediated exocytosis into the synaptic cleft. This cleft is home to acetylcholinesterase, the enzyme that breaks down ACh. Fifty percent of the released ACh is cleaved by acetylcholinesterase, and the remaining molecules bind to postsynaptic nAChRs on the motor end plate, which causes ion channels to open. When enough channels are opened, the myocyte is depolarized and muscle contraction occurs. Acetylcholine remaining in the synapse is rapidly degraded by acetylcholinesterase, and the muscle is allowed to repolarize.

SUCCINYLCHOLINE

[Succinylcholine](#) (also known as diacetylcholine or suxamethonium chloride) is the only depolarizing NMBA available for clinical use.

Pharmacology of succinylcholine — [Succinylcholine](#) produces a reliable neuromuscular blockade that is the fastest in onset and shortest in duration of all NMBAs. Thus succinylcholine is often used for rapid sequence induction and intubation. (See "[Rapid sequence induction and intubation \(RSII\) for anesthesia](#)", [section on 'Succinylcholine'](#) and '[Endotracheal intubation](#)' above.)

Within 30 seconds after intravenous (IV) administration, patients experience fasciculations from depolarization and antidromic activation of unparalyzed portions of the motor unit [125]; flaccid paralysis ensues seconds later. Fasciculations of varying duration and intensity occur in 94 percent of patients [126]. The resultant depolarizing block is often referred to as a Phase I block in which there is no fade following TOF stimulation [127]. While Phase I block has been classically associated with [succinylcholine](#), the use of a quantitative monitor following succinylcholine administration reveals that Phase I block does not typically occur in routine clinical practice [128].

- **Dosing/onset of action** – An IV dose of 1 to 1.5 mg/kg, or 3 to 5 x ED₉₅, yields flaccid paralysis in one to two minutes. After an intubating dose, the clinical duration of action (defined as the time from drug administration until recovery of single twitch to 25 percent of baseline)

is 7 to 12 minutes. If IV access is unavailable, [succinylcholine](#) can also be administered intramuscularly (3 to 4 mg/kg), and paralysis will occur in approximately four minutes.

[Succinylcholine](#) is occasionally administered by infusion, titrated to twitch depression, most often when a brief period of neuromuscular blockade is required (such as electroconvulsive therapy). Use of a succinylcholine infusion or by repeated bolus administration increase the risk of Phase II block, and prolonged paralysis. Phase II block usually occurs after total succinylcholine doses exceeding 4 mg/kg. Alternatives to succinylcholine infusion for short procedures include mivacurium (where available), or intermediate duration steroidal NMBA (ie, [rocuronium](#) or [vecuronium](#)) followed by reversal with [sugammadex](#). (See '[Mivacurium](#)' below and '[Sugammadex](#)' above.)

- **Pharmacokinetics** – [Succinylcholine](#) is metabolized by butyrylcholinesterase (also known as pseudocholinesterase or plasma cholinesterase) in a process that yields no active metabolites.

Phase II block — Phase II block typically develops after large doses of [succinylcholine](#) (>4 mg/kg), after repeated doses, or after a continuous infusion; some characteristics of phase II block have been described in doses as low as 0.3 mg/kg [129]. This type of block occurs when the post-junctional membrane action potential eventually returns to baseline, despite the continued activation of nAChRs in the presence of succinylcholine. Phase II block is associated with some of the features of nondepolarizing block, including train-of-four (TOF) fade and post-tetanic potentiation.

Butyrylcholinesterase (pseudocholinesterase) deficiency — In patients with atypical or deficient butyrylcholinesterase, recovery from [succinylcholine](#) (and mivacurium) can be prolonged. A variety of factors can reduce butyrylcholinesterase activity but only modestly prolong paralysis with succinylcholine, including liver disease, pregnancy, advanced age, kidney failure, major burns, oral contraceptives, and [echothiophate](#) eye drops. In contrast, abnormal genetic variants of the butyrylcholinesterase enzyme can cause marked prolongation of paralysis after succinylcholine administration. Duration of paralysis may be increased by 50 to 100 percent in patients who are heterozygous for abnormal butyrylcholinesterase. Patients who are homozygous for the abnormal gene may be paralyzed for up to eight hours after an intubating dose of succinylcholine, and may require prolonged postoperative mechanical ventilation [130].

Atypical butyrylcholinesterase is often diagnosed after a patient has an unexpectedly prolonged response to [succinylcholine](#). Objective TOF monitoring should be used, and recovery documented prior to extubation, for all patients who have received NMBA, including those who have received succinylcholine. Some degree of recovery from succinylcholine-induced block should be documented prior to administration of a nondepolarizing NMBA, as administration of a nondepolarizing NMBA prior to documenting recovery from succinylcholine will complicate evaluation and subsequent management, should prolonged succinylcholine block occur. Principles for management of prolonged block after succinylcholine should include the following:

- Causes for prolonged block other than butyrylcholinesterase deficiency should be ruled out, including hypothermia and electrolyte disorders, or malfunctioning of the peripheral nerve stimulator or neuromuscular monitor.
- TOF fade is indicative of phase II block that is likely to occur in these patients. Reversal with [neostigmine](#) should **not** be attempted. Anticholinesterase agents can worsen paralysis and induce phase II block in this setting. Lack of fade may indicate resolution of phase II block, but the degree of recovery from phase I block cannot be assessed without a baseline, pre-NMBA twitch.
- The patient should be sedated and mechanically ventilated until full return of muscle strength as determined by quantitative monitoring if available. If quantitative monitoring is not available, the subjective monitoring with a peripheral nerve stimulator is warranted to confirm recovery.

After an episode of prolonged paralysis after [succinylcholine](#), testing may be performed for the level and activity of butyrylcholinesterase. These tests are not routinely performed in hospital laboratories, and blood samples are usually sent to specialized testing centers. Two tests are performed, one for the level of enzyme activity, and if enough activity exists, another test is performed for the [dibucaine](#) number, which indicates the percent of inhibition of the enzyme under standardized test conditions. The blood sample for these tests should be drawn after the patient has recovered full muscle strength, since the enzyme activity results can vary depending on anesthetic exposure [131]. Dibucaine inhibition (ie, phenotype testing) is not affected by anesthetic exposure.

[Dibucaine](#) inhibition of 70 to 80 percent indicates a normal enzyme, whereas dibucaine inhibition of 20 to 30 percent indicates a homozygous atypical enzyme.

Since atypical butyrylcholinesterase (ie, pseudocholinesterase deficiency) is inherited in an autosomal recessive pattern [132], questions may arise for patients reporting prolonged paralysis after [succinylcholine](#) administration in a first-degree relative.

Management options may include:

- Not using [succinylcholine](#). If rapid sequence induction and intubation is necessary, high-dose [rocuronium](#) could be used with the use of [sugammadex](#) for reversal. (See '[Sugammadex](#)' above.)
- For preoperative testing, the [dibucaine](#) number may be checked, as discussed above.
- If available, testing for plasma pseudocholinesterase levels can be performed to confirm atypical butyrylcholinesterase.

Adverse effects of succinylcholine — [Succinylcholine](#) is associated with a number of adverse effects, many of which are related to its depolarizing effect. Succinylcholine is contraindicated for patients with susceptibility to malignant hyperthermia, denervating conditions, major burns after 48 hours, and severe hyperkalemia.

- **Hyperkalemia** – Serum potassium levels increase by approximately 0.5 mEq/L after the administration of 1 mg/kg [succinylcholine](#). This increase is exaggerated and may be life-threatening in patients who have upregulation of extrajunctional nicotinic acetylcholine receptors (nAChRs; eg, stroke with paralysis, spinal cord injury, muscle and muscle membrane disorders, muscular sclerosis, and polyneuropathies, sepsis, prolonged bed rest), or significant burns [133,134]. The administration of a small dose of a nondepolarizing NMBA prior to succinylcholine (ie, a defasciculating dose) does not reliably prevent the rise in potassium associated with succinylcholine.

Administration of [succinylcholine](#) for routine intubation in children has largely been abandoned. The Food and Drug Administration of the United States has issued a boxed warning for succinylcholine for children, except for emergency airway management, over concerns for acute rhabdomyolysis and hyperkalemia in children with undiagnosed muscular dystrophies. Despite the warning, succinylcholine can be used in carefully screened children in select circumstances (eg, for rapid sequence induction and intubation [RSII]). (See "[General anesthesia in neonates and children: Agents and techniques](#)", section on '[Intravenous induction medications](#)'.)

- **Malignant hyperthermia** – [Succinylcholine](#) is a known trigger for malignant hyperthermia (MH), and its use is contraindicated in patients at risk for MH. (See "[Malignant hyperthermia susceptibility: Evaluation](#)" and "[Malignant hyperthermia: Diagnosis and management of acute crisis](#)".)
- **Myalgias** – Nearly one-half of patients experience myalgias after receiving [succinylcholine](#). Such myalgias were initially thought to be related to the fasciculations. However, the use of a nondepolarizing NMBA or "defasciculating" dose may attenuate, but does not reliably prevent, myalgias [15]. Nonsteroidal anti-inflammatory drugs are the most effective means of attenuating succinylcholine-induced myalgias (number-needed-to-treat of 2.5 to 3), but only prevent myalgia in one-third of patients [126].

Pretreatment with nondepolarizing NMBAs can produce partial neuromuscular blockade, which results in visual disturbances, difficulty breathing, and other adverse effects [126].

- **Increased intragastric pressure** – While [succinylcholine](#) increases intragastric pressure, the risk of aspiration is mitigated by a simultaneous increase in the lower esophageal sphincter tone.
- **Increased intraocular pressure** – [Succinylcholine](#) can cause increased intraocular pressure by up to 15 mmHg via unknown mechanisms. The optimal strategy for RSII in the setting of an open eye with a full stomach is debated, and must be individualized. (See "[Anesthesia for emergency eye surgery](#)", section on '[Induction of anesthesia](#)'.)
- **Increased intracranial pressure** – [Succinylcholine](#) can produce a transient increase in intracranial pressure, which can be attenuated by administration of a defasciculating dose of nondepolarizing NMBA. (See "[Anesthesia for craniotomy in adults](#)", section on '[Neuromuscular blocking agents](#)'.)
- **Cardiac dysrhythmias** – [Succinylcholine](#) may cause a variety of arrhythmias [135]. Bradycardia is the most common of these, and occurs most frequently in children, or with repeated administration. Some clinicians routinely administer [atropine](#) prior to RSII with succinylcholine in children to prevent bradycardia or asystole. (See "[General anesthesia in neonates and children: Agents and techniques](#)", section on '[Rapid sequence induction and intubation](#)'.)
- **Allergic reactions** – NMBAs are among the commonly implicated triggers for perioperative anaphylaxis. The reported incidence of anaphylaxis varies, but it appears that the rates of anaphylaxis to [succinylcholine](#) and [rocuronium](#) are significantly higher than the rates of reaction to other NMBAs [136]. (See "[Perioperative anaphylaxis: Allergy evaluation and prevention of recurrent reactions](#)", section on '[Neuromuscular-blocking agents](#)'.)

NONDEPOLARIZING NEUROMUSCULAR BLOCKING AGENTS

Nondepolarizing, competitive NMBAs are classified by their chemical structure (steroidal versus benzylisoquinolinium) as well as their duration of action (short-, intermediate-, or long-acting agents). The properties of NMBAs are shown in tables ([table 4](#) and [table 3](#) and [table 5](#)).

Pharmacology of nondepolarizing neuromuscular blocking agents — In general, the speed of onset of neuromuscular block depends on the potency of the drug; greater potency is associated with slower onset of block. Thus, [rocuronium](#), with an ED₉₅ of 0.3 mg/kg IV, has a much more rapid onset than [vecuronium](#), which has an ED₉₅ of 0.05 mg/kg ([table 4](#)). The standard intubating dose of nondepolarizing NMBAs is 2 to 3 times the ED₉₅ for the specific drug. Higher doses of some drugs speed onset of block and may be used for rapid sequence induction and intubation (RSII), though with a prolonged duration of block.

Properties of neuromuscular blocking agents									
Agent	Chemical structure	Onset (min)	Duration (min)	ED ₉₅ (mg/kg IV)	Intubating dose (mg/kg IV)	Reversal agent	Contraindications	Adverse effects	Comments
Succinylcholine	Depolarizing	30-60	3-5	0.1	1-2	None	Malignant hyperthermia, hyperkalemia, burns, neuromuscular disorders	Myalgias, hyperkalemia, increased IOP, increased IGP	Used for rapid sequence induction
Rocuronium	Nondepolarizing	1-3	30-60	0.3	1-1.5	Neostigmine, Sugammadex	None	Bradycardia, hypotension	Standard agent for RSII
Vecuronium	Nondepolarizing	2-4	30-60	0.05	0.1-0.2	Neostigmine, Sugammadex	None	Bradycardia, hypotension	Standard agent for RSII
Pipecuronium	Nondepolarizing	2-4	30-60	0.05	0.1-0.2	Neostigmine, Sugammadex	None	Bradycardia, hypotension	Standard agent for RSII
Doxacurium	Nondepolarizing	2-4	30-60	0.05	0.1-0.2	Neostigmine, Sugammadex	None	Bradycardia, hypotension	Standard agent for RSII
Atracurium	Nondepolarizing	2-4	30-60	0.05	0.1-0.2	Neostigmine, Sugammadex	None	Bradycardia, hypotension	Standard agent for RSII
Cisatracurium	Nondepolarizing	2-4	30-60	0.05	0.1-0.2	Neostigmine, Sugammadex	None	Bradycardia, hypotension	Standard agent for RSII
Benziisoquinolinium	Nondepolarizing	2-4	30-60	0.05	0.1-0.2	Neostigmine, Sugammadex	None	Bradycardia, hypotension	Standard agent for RSII
Steroidal	Nondepolarizing	2-4	30-60	0.05	0.1-0.2	Neostigmine, Sugammadex	None	Bradycardia, hypotension	Standard agent for RSII

Properties of neuromuscular blocking agents

A dose of 10 percent of the ED₉₅ may be used to maintain an existing level of neuromuscular block during anesthesia, guided by neuromuscular monitoring. (See 'Facilitation of surgery' above.)

Table 5 - larger image below

- **Steroidal compounds** – [Rocuronium](#) and [vecuronium](#) are intermediate-acting, steroidal NMBAs. Pancuronium is the only available long-acting steroidal NMBA; it is no longer available in North America, Europe, and most areas around the world.
- **Benzylisoquinolinium compounds** – Mivacurium is the only short-acting, benzylisoquinolinium NMBA. It is no longer available in the United States, but is available in some other countries.

Pharmacokinetic properties of aminosteroid nondepolarizing neuromuscular blocking agents in infants, children, and adults*

	Rocuronium				Vecuronium			
	Infant	Child	Adult	Elderly	Infant	Child	Adult	Elderly
ED ₉₅ (mg/kg)	0.3	0.4	0.3	0.3	0.05	0.08	0.05	0.05
V _d (L/kg)	0.2	0.17	0.3 to 0.2 to 0.6	0.2 to 0.6	0.25	0.25	0.4	0.3
T _{1/2β} (minutes)	50 to 110	25 to 100	100 to 200	100 to 200	60	40 to 60	70 to 100	70 to 100
Onset time (minutes)	0.6	1.3	1.5 to 2 to 5	2 to 5	1.5	2.4	3 to 4	3 to 4
Recovery index (minutes)	40 to 100	27 to 70	30 to 70	30 to 70	30 to 70	25 to 50	40 to 90	40 to 90

Pharmacokinetic properties of aminosteroid nondepolarizing neuromuscular blocking agents in infants, children, and adults*

Table 4 - larger image below

Steroidal neuromuscular blocking agents

Rocuronium — [Rocuronium](#) has a faster onset than other nondepolarizing NMBAs. Thus rocuronium may be used at higher doses as an alternative to [succinylcholine](#) for RSII. (See "Induction agents for rapid sequence intubation in adults for emergency medicine and critical care".)

[Rocuronium](#) does not release histamine, nor is it vagolytic, and it causes minimal hemodynamic effects. The incidence of allergic reactions to rocuronium and [succinylcholine](#) appears to be higher than other NMBAs. (See "Perioperative anaphylaxis: Allergy evaluation and prevention of recurrent reactions", section on 'Neuromuscular-blocking agents'.)

- **Dosing/onset of action** – The standard intubating dose of 0.6 mg/kg (2 x ED₉₅) provides adequate intubating conditions in 1.5 to 3 minutes in most (but not all) patients. (See 'Endotracheal intubation' above.)

After an intubating dose, the clinical duration of action is 30 to 70 minutes. Thereafter, doses of 0.1 mg/kg or an infusion at a dose of 5 to 12 mcg/kg/minute can be used to maintain neuromuscular blockade, guided by neuromuscular monitoring. The RSII dose of [rocuronium](#) of 1.2 mg/kg, or 4 x ED₉₅, has a similar average onset time as [succinylcholine](#), although the variability in onset is greater with rocuronium [137].

- **Pharmacokinetics** – [Rocuronium](#) has a volume of distribution of 0.3 to 0.7 L/kg. It is excreted mostly through the biliary route, although a small portion is excreted renally. Its metabolism results in 17-desacetyl-rocuronium, a compound that has 20 percent the activity of the parent compound.

Vecuronium — [Vecuronium](#) is a steroidal NMBA with minimal hemodynamic effects. It is not used as an alternative to [succinylcholine](#) for RSII because of slower onset time compared with [rocuronium](#).

- **Dosing/onset of action** – An intubating dose of [vecuronium](#) of 0.1 mg/kg, or 2 x ED₉₅ provides adequate intubating conditions in three to four minutes. The maintenance dose is 0.01 mg/kg, or an infusion at 1 to 2 mcg/kg/minute, guided by neuromuscular monitoring. Following an intubating dose, the clinical duration of action is 25 to 50 minutes.
- **Pharmacokinetics** – [Vecuronium](#) has a volume of distribution of 0.4 L/kg. It is excreted both through the biliary and renal routes. Thus patients with renal or hepatic disease may have a prolonged response to vecuronium, and maintenance doses should be carefully guided by neuromuscular monitoring. Vecuronium is metabolized to 3-desacetyl-vecuronium, a compound that has 60 percent the activity of the parent compound.

Benzylisoquinolinium neuromuscular blocking agents

Atracurium — [Atracurium](#) is an intermediate-acting benzylisoquinolinium NMBA that is a mixture of 10 isomers ([table 3](#)).

- **Dosing/onset of action** – An intubating dose of [atracurium](#) of 0.5 mg/kg, or 2 x ED₉₅ provides adequate intubating conditions in three to five minutes. The maintenance dose of atracurium is 0.1 mg/kg, or an infusion at 10 to 20 mcg/kg/minute, guided by neuromuscular monitoring. At doses higher than 0.5 mg/kg, atracurium increases plasma histamine levels, causing skin flushing, hypotension, and tachycardia [138]. Following an intubating dose, the clinical duration of atracurium is 30 to 45 minutes.
- **Pharmacokinetics** – [Atracurium](#) is metabolized through nonspecific plasma esterase-mediated hydrolysis and a nonenzymatic, pH- and temperature-dependent degradation called Hofmann elimination. Laudanosine is the final product of Hofmann elimination, a compound that was found to be epileptogenic in animal studies when administered in large doses [139]. Epileptogenic side effects have not been demonstrated in humans, and this byproduct has no significance at clinically relevant doses [140]. Metabolism is essentially independent of hepatic and kidney function, and atracurium has no active metabolites. The volume of distribution of atracurium is 0.15 L/kg.

Cisatracurium — [Cisatracurium](#) is the *cis*-isomer of [atracurium](#) and is four times more potent than atracurium. In contrast with atracurium, it does not cause histamine release ([table 3](#)).

- **Dosing/onset of action** – The intubating dose of [cisatracurium](#) of 0.15 to 0.2 mg/kg, or 3 x ED₉₅, provides adequate intubating conditions in four to seven minutes. The maintenance dose of cisatracurium is 0.01 mg/kg or an infusion at 1 to 3 mcg/kg/min, guided by neuromuscular monitoring. Following an intubating dose of cisatracurium, the clinical duration of action is 35 to 50 minutes.
- **Pharmacokinetics** – [Cisatracurium](#) is also primarily metabolized through Hofmann elimination. Cisatracurium has no active metabolites, and its volume of distribution is 0.16 L/kg.

Mivacurium — Mivacurium is a short-acting nondepolarizing benzylisoquinolinium NMBA that was developed as a nondepolarizing alternative to [succinylcholine](#). Large doses are required for intubation because of this drug's low potency. Clinical utility has been limited by the histamine release associated with these high doses [138]. Mivacurium is no longer available in the United States or Canada but is available in many other countries.

- **Dosing/onset of action** – An intubating dose of mivacurium of 0.2 mg/kg, or 3 x ED₉₅, provides adequate intubating conditions in three to four minutes. The maintenance dose of mivacurium is 0.1 mg/kg [141], or an infusion at 5 to 8 mcg/kg/min, guided by neuromuscular monitoring. Following an intubating dose of mivacurium, the clinical duration of action is 15 to 20 minutes.
- **Pharmacokinetics** – Mivacurium is mostly metabolized by butyrylcholinesterase and should not be used in patients with butyrylcholinesterase deficiency, as such patients will have unpredictable and prolonged response. Mivacurium has a volume of distribution of 0.2 L/kg. It does not have active metabolites.

Reversal (antagonism) of mivacurium with either [neostigmine](#) or edrophonium is faster than spontaneous recovery [142], although neostigmine reversal of profound mivacurium block may result in prolongation of the block [143].

Neuromuscular block that occurs after mivacurium administration in patients with pseudocholinesterase deficiency is not completely reversed by [neostigmine](#) [144], and recovery from neuromuscular block in this setting should be confirmed by objective monitoring, with a train-of-four ratio (TOFR) >0.9.

SUMMARY AND RECOMMENDATIONS

- **Use and selection of neuromuscular blocking agents (NMBAs)** – NMBAs are administered during anesthesia to facilitate endotracheal intubation and/or to improve surgical conditions ([table 6](#)). Selection of NMBA is based on patient factors and the clinical indication. (See '[Selection of neuromuscular blocking agents](#)' above.)

Patient factors can affect the response to NMBAs, including neuromuscular disease ([table 1](#)), burns, extremes of age, hepatic and kidney dysfunction, and physiologic disturbances. Other medications (eg, inhaled anesthetics) can also affect the response to NMBAs. (See '[Factors that affect the response to neuromuscular blocking agents](#)' above.)

- **Depth of block** – Deep neuromuscular block (ie, 0 to 1 twitch, post tetanic count ≥ 1 with neuromuscular monitoring) is used for endotracheal intubation, and for some surgical indications. We aim for a moderate level of neuromuscular block (ie, 1 to 3 twitches) for most surgical procedures. (See '[Clinical use of neuromuscular blocking agents](#)' above.)
- **Avoiding residual block** – Full recovery of neuromuscular function (ie, train-of-four [TOF] ratio of ≥ 0.9) must be achieved prior to extubation of the trachea. Residual neuromuscular block is associated with weakness of upper airway muscles, increased risk of aspiration, hypoxemia, awareness during emergence from anesthesia, and increased postoperative pulmonary complications. (See '[Avoidance of residual neuromuscular blockade](#)' above.)

A strategy for avoidance of residual neuromuscular blockade should include (see '[Avoidance of residual neuromuscular blockade](#)' above):

- Use intermediate or short acting NMBAs whenever possible
- Avoid deep neuromuscular block (ie, train-of-four count [TOFC] of zero) when clinically appropriate
- Use objective neuromuscular monitoring whenever possible to guide administration of NMBAs and assess recovery
- Administer reversal agents in appropriate doses, guided by the degree of recovery from neuromuscular block. (See '[Reversal of neuromuscular block](#)' above.)
- Extubate the trachea only after a TOF ratio ≥ 0.9 is achieved, as indicated by objective monitoring
- If a quantitative monitor is not available, we suggest administering [neostigmine](#) only when TOFC = 4 (as assessed at the adductor pollicis muscle) ([Grade 2C](#)). We allow at least 10 minutes after neostigmine administration before tracheal extubation. (See '[Anticholinesterases](#)' above.)
- **Reversal of block** – Reversal of neuromuscular blockade can occur by spontaneous recovery, or by administration of reversal agents (ie, [neostigmine](#) or [sugammadex](#)). Sugammadex can only be used to reverse block after administration of aminosteroid NMBAs (eg, [rocuronium](#) or [vecuronium](#)). For patients who have received rocuronium or vecuronium and who require reversal of block, we suggest

using sugammadex rather than neostigmine (**Grade 2C**). Sugammadex reverses block more quickly and reliably than neostigmine, can be used to reverse deeper levels of neuromuscular block, and may be associated with reduced incidence of postoperative pulmonary complications (PPCs). (See '[Sugammadex versus neostigmine](#)' above.)

Where cost is an issue, it is reasonable to routinely use [neostigmine](#) and restrict the use of [sugammadex](#) to patients who are at higher risk of complications of postoperative residual neuromuscular block (eg, patients >80 years of age or who undergo thoracic surgery, and/or patients who need reversal from deep levels of neuromuscular blockade (TOFC ≤ 3).

The doses of [neostigmine](#) and [sugammadex](#) should be based on the level of existing neuromuscular block at the time of administration ([table 6](#)). (See '[Reversal of neuromuscular block](#)' above.)

- **Classification of NMBAs** – NMBAs are classified according to their interaction with nicotinic acetylcholine receptors (nAChRs) at the neuromuscular junction (NMJ). [Succinylcholine](#) produces skeletal muscle relaxation by binding directly with nAChRs to cause prolonged depolarization. Nondepolarizing NMBAs act as competitive antagonists for acetylcholine at nAChRs to prevent the initiation of an action potential ([table 5](#)). (See '[Classification of neuromuscular blocking agents](#)' above.)
- **Succinylcholine (depolarizer)** – [Succinylcholine](#) is most often used for endotracheal intubation. It causes muscle fasciculations, an increase in serum potassium, and may cause postoperative myalgias. Succinylcholine is contraindicated in patients with severe burns, denervating neuromuscular diseases and stroke, susceptibility to malignant hyperthermia, and butyrylcholinesterase deficiency. (See '[Succinylcholine](#)' above.)
- **Nondepolarizers**
 - [Rocuronium](#) and [vecuronium](#) are the commonly used intermediate duration aminosteroid nondepolarizing NMBAs. They are used for endotracheal intubation and to facilitate surgery. Vecuronium and rocuronium have minimal cardiovascular side effects. They should be avoided or used with careful monitoring in patients with renal or hepatic dysfunction ([table 4](#)). (See '[Steroidal neuromuscular blocking agents](#)' above.)
 - [Cisatracurium](#), [atracurium](#), and mivacurium are benzylisoquinolinium nondepolarizing NMBAs. Cisatracurium and atracurium are intermediate duration of action NMBAs that are used for endotracheal intubation and for facilitation of surgery. They are metabolized through Hofmann elimination, and do not depend on renal or hepatic function. Atracurium, but not cisatracurium, releases histamine at higher doses and can cause hypotension ([table 3](#)). (See '[Atracurium](#)' above and '[Cisatracurium](#)' above.)
 - Mivacurium is a short acting NMBA that is metabolized by plasma butyrylcholinesterase, and should be avoided in patients with atypical or deficient butyrylcholinesterase. At high doses, mivacurium releases histamine ([table 5](#)). (See '[Mivacurium](#)' above.)

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GRAPHICS

Table 1: Sensitivity of patients with neuromuscular disease to neuromuscular blocking agents

Disorder type	Response to succinylcholine	Response to nondepolarizing NMBA
Myasthenic syndromes		
▪ Myasthenia gravis	Resistant	Sensitive
▪ Lambert-Eaton myasthenic syndrome	Sensitive	Sensitive
▪ Congenital myasthenic syndromes	Variable, not recommended due to potential hyperkalemic response	Sensitive
Malignant hyperthermia, central core disease, King-Denborough syndrome	Contraindicated	Normal
Demyelinating diseases		
▪ Multiple sclerosis	Avoid use	Sensitive
▪ Motor neuron diseases	Avoid use	Sensitive
▪ Guillain-Barré syndrome	Avoid use	Sensitive
▪ Charcot-Marie-Tooth disease (CMTD)	Avoid use	Normal
Primary muscle diseases		
Muscular dystrophies	Contraindicated	Sensitive
• Duchenne muscular dystrophy	Contraindicated	Sensitive
• Becker muscular dystrophy	Contraindicated	Sensitive
• Limb-girdle dystrophy	Contraindicated	Sensitive
• Congenital muscular dystrophy	Contraindicated	Sensitive
• Facioscapulohumeral muscular dystrophy	Contraindicated	Sensitive
• Myotonic dystrophy	Contraindicated	Sensitive*
Mitochondrial myopathies	Sensitive	Sensitive
Motor neuron diseases		
Amyotrophic lateral sclerosis (Lou Gehrig's disease)	Avoid use	Sensitive
Channelopathies		
▪ Acquired neuromyotonia	Avoid use	Normal*
▪ Myotonia congenita	Avoid use	
▪ Paramyotonia congenita	Avoid use	
▪ Potassium aggravated myotonia	Avoid use	
▪ Hyperkalemic periodic paralysis	Avoid use	
▪ Hypokalemic periodic paralysis	Avoid use	
Spinal cord injury, stroke, immobility, ICU patients	Avoid use	Resistant
Metabolic myopathies		
Storage disorders (lipid, glycogen) [¶]	Avoid use	Variable – avoid use if possible

NMBA: neuromuscular blocking agent; ICU: intensive care unit.

* Anticholinesterase drugs should be avoided because they may provoke a myotonic reaction.

[¶] Carnitine palmitoyltransferase II (CPT II) deficiency; myophosphorylase deficiency (McArdle disease).

Adapted from: Brull SJ. Neuromuscular blocking agents. In: Clinical Anesthesia, 8th ed, Barash PG, Cullen BF, Stoelting RK, et al (Eds), Wolters Kluwer 2017. p.527.

Table 2: Conditions associated with malignant hyperthermia or rhabdomyolysis related to anesthesia

Condition	Reported problems with anesthesia	Genes potentially related to MH or anesthesia induced rhabdomyolysis	Recommendations for anesthesia	
			Volatile anesthetics	Succinylcholine
Conditions associated with MH				
MHS, no associated syndrome	MH reactions	<i>RYR1</i> <i>CACNA1S</i> <i>STAC3</i>	Contraindicated	Contraindicated
Exertional rhabdomyolysis	MH in some patients with <i>RYR1</i> pathogenic variants	<i>RYR1</i>	Contraindicated in patients with <i>RYR1</i> pathogenic variants or rhabdomyolysis in unexpected situations*	Contraindicated in patients with <i>RYR1</i> pathogenic variants or rhabdomyolysis in unexpected situations*
Myopathies				
Central core myopathy	High incidence of MH in patients with <i>RYR1</i> receptor abnormality	<i>RYR1</i>	Contraindicated	Contraindicated
Multiminicore myopathy	High incidence of MH in patients with biallelic <i>RYR1</i> pathogenic variants	<i>RYR1</i>	Contraindicated	Contraindicated
King-Denborough syndrome	High incidence of MH in patients with <i>RYR1</i> receptor abnormality	<i>RYR1</i>	Contraindicated	Contraindicated
STAC3 (Native American) myopathy	High incidence of MH in patients homozygous for <i>STAC3</i> pathogenic variant	<i>STAC3</i>	Contraindicated	Contraindicated
Congenital myopathy with cores and rods	MH in patient with multiple <i>RYR1</i> pathogenic variants	<i>RYR1</i>	Contraindicated in patients with <i>RYR1</i> pathogenic variants*	Contraindicated
Nemaline myopathy	Uncomplicated use of succinylcholine and potent inhalational anesthesia has been reported, however there is insufficient evidence to support safety of succinylcholine	<i>RYR1</i>	Contraindicated in patients with <i>RYR1</i> pathogenic variants*	Contraindicated
Centronuclear myopathy (CNM)	MH episodes likely only in patients with <i>RYR1</i> pathogenic variants (and possibly <i>CACNA1S</i> pathogenic variants, though these patients also typically have cores) and not in other subtypes	<i>RYR1</i> <i>CACNA1S</i>	Contraindicated in patients with <i>RYR1</i> pathogenic variants*	Contraindicated in patients with <i>RYR1</i> pathogenic variants*
Congenital fiber type disproportion (CFTD)	No reports of MH; 10 to 20% of patients with CFTD may have <i>RYR1</i> pathogenic variants	<i>RYR1</i> <i>CACNA1S</i>	Contraindicated in patients with <i>RYR1</i> pathogenic variants*	Contraindicated in patients with <i>RYR1</i> pathogenic variants*
Periodic paralysis	MH in patients with <i>RYR1</i> pathogenic variant MHS has not been associated with <i>CACNA1S</i> -linked hypokalemic periodic paralysis Atypical normokalemic periodic paralysis has been reported in a small number of patients with <i>RYR1</i> pathogenic variants	<i>RYR1</i> <i>CACNA1S</i>	Contraindicated in patients with <i>RYR1</i> pathogenic variants*	Contraindicated in patients with <i>RYR1</i> pathogenic variants*
HyperCKemia	MH in patients with <i>RYR1</i> pathogenic variants	<i>RYR1</i>	Contraindicated in patients with <i>RYR1</i> pathogenic variants*	Contraindicated in patients with <i>RYR1</i> pathogenic variants*
Conditions associated with rhabdomyolysis				

Exertional rhabdomyolysis	MH in some patients with <i>RYR1</i> pathogenic variants	<i>RYR1</i>	Contraindicated in patients with <i>RYR1</i> pathogenic variants or rhabdomyolysis in unexpected situations*	Contraindicated in patients with <i>RYR1</i> pathogenic variants or rhabdomyolysis in unexpected situations*
Dystrophinopathies	Patients with <i>DMD</i> pathogenic variants: Severe hyperkalemia with succinylcholine; possible rhabdomyolysis or hyperkalemic cardiac arrest following volatile anesthetics	<i>DMD</i>	Only if essential, in limited amounts, eg, for inhalational induction to establish venous access	Contraindicated
Myoadenylate deaminase deficiency	Rhabdomyolysis in patients homozygous for <i>AMPD1</i> pathogenic variant	<i>AMPD1</i>	Try to avoid in individuals homozygous for <i>AMPD1</i> pathogenic variant	Try to avoid
Myophosphorylase deficiency (McArdle disease, glycogen storage disease V)	In patients with <i>PYGM</i> pathogenic variant succinylcholine may exacerbate underlying symptoms	<i>PYGM</i>	Not contraindicated	Try to avoid Avoid in patients with <i>PYGM</i> pathogenic variant*
Carnitine palmitoyl transferase type 2 deficiency	Perioperative rhabdomyolysis in patients with <i>CPT2</i> pathogenic variants	<i>CPT2</i>	Not contraindicated; no evidence of superior safety with propofol	Try to avoid Avoid in patients with <i>CPT2</i> pathogenic variant*
Other conditions				
Osteogenesis imperfecta	Some reports of fever with anesthesia		OK to use	OK to use
Arthrogryposis multiplex congenita	Some reports of fever with anesthesia Most cases have neurologic, not myopathic etiology	<i>RYR1</i> , rare Most cases have neurologic, not myopathic etiology	OK to use in absence of <i>RYR1</i> pathogenic variants	Avoid if suspected or confirmed myopathic etiology
Myotonia	Prolonged muscle rigidity after succinylcholine Possible rigidity after anticholinesterase (NMBA reversal)		OK to use	Avoid succinylcholine or anticholinesterase reversal agents
Neuroleptic malignant syndrome	Idiosyncratic reaction to antidopaminergics (eg, haloperidol, droperidol), usually more than 24 hours after administration		OK to use	Contraindicated in patients with rhabdomyolysis OK to use in patients without rhabdomyolysis

MH: malignant hyperthermia; MHS: malignant hyperthermia susceptible; *RYR1*: ryanodine receptor 1; *CACNA1S*: calcium voltage-gated channel subunit alpha1 S; *STAC3*: SH3 and cysteine-rich domains 3; *DMD*: dystrophin; *AMPD1*: Adenosine monophosphate deaminase-1; *PYGM*: phosphorylase, glycogen, muscle; *CPT2*: carnitine palmitoyltransferase 2.

* This recommendation is applicable to patients with the specified pathogenic variant and to patients with unknown genetic test results.

Table 3: Pharmacokinetic properties of benzylisoquinolinium nondepolarizing neuromuscular blocking agents in infants, children, and adults*

	Atracurium				Cisatracurium			
	Infant	Child	Adult	Elderly	Infant	Child	Adult	Elderly
ED ₉₅ (mg/kg)	0.16 to 0.20	0.25	0.25	0.25	0.04	0.05	0.05	0.05
V _D (L/kg)	0.21	0.13	0.14	0.18	NA	0.21 to 0.35	0.12 to 0.26	0.14 to 0.26
t _{½-Beta} (minutes)	14 to 20	14 to 20	21	22	NA	21 to 23	23 to 30	21 to 29
Onset time (minutes)	2 to 3	2 to 3	3 to 5	3 to 5	2.5	2.5	4 to 7	5 to 7
Dosing interval (minutes)	22 to 29	28	30 to 45	30 to 45	42	36	35 to 50	35 to 50

Onset times and dosing intervals are averages for a 2 × ED₉₅ dose. Infant: 0 to 1 year; Child: 1 to 18 years; Adult: 18 to 65 years; Elderly: >65 years.

ED₉₅: effective dose required to achieve 95% depression in evoked twitch response; V_D: volume of distribution; t_{½-Beta}: elimination half-life; NA: data not available.
* The data are averages obtained from published literature and do not account for other influences such as volatile anesthetics, organ dysfunction, etc.

Adapted from: Brull SJ. Neuromuscular blocking agents. In: Clinical Anesthesia, 8th ed, Barash PG, Cullen BF, Stoelting RK, et al (Eds), Wolters Kluwer 2017. p.527.

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Table 4: Pharmacokinetic properties of aminosteroid nondepolarizing neuromuscular blocking agents in infants, children, and adults*

	Rocuronium				Vecuronium			
	Infant	Child	Adult	Elderly	Infant	Child	Adult	Elderly
ED ₉₅ (mg/kg)	0.3	0.4	0.3	0.3	0.05	0.08	0.05	0.05
V _D (L/kg)	0.2	0.17	0.3 to 0.7	0.2 to 0.6	0.35	0.20	0.4	0.3
t _{½-Beta} (minutes)	50 to 110	25 to 110	100 to 250	110 to 280	65	41.0	65 to 75	70 to 180
Onset time (minutes)	0.6	1.3	1.5 to 3	2 to 5	1.5	2.4	3 to 4	3 to 4
Dosing interval (minutes)	42 to 100	27	30 to 70	22 to 73	59	37	25 to 50	49 to 97

Onset times and dosing intervals are averages for a 2 × ED₉₅ dose. Infant: 0 to 1 year; Child: 1 to 18 years; Adult: 18 to 65 years; Elderly: >65 years.

ED₉₅: effective dose required to achieve 95% depression in evoked twitch response; V_D: volume of distribution; t_{½-Beta}: elimination half-life.
* The data are averages obtained from published literature and do not account for other influences such as volatile anesthetics, organ dysfunction, etc.

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Table 5: Properties of neuromuscular blocking agents

Agent	Vecuronium	Rocuronium	Pancuronium*	Mivacurium [¶]	Atracurium	Cisatracurium	Succinylcholine
Type (structure)	Non-depolarizing	Non-depolarizing	Non-depolarizing	Non-depolarizing	Non-depolarizing	Non-depolarizing	Depolarizing
Type (duration)	Intermediate	Intermediate	Long	Short	Intermediate	Intermediate	Ultrashort
Potency – ED ₉₅ (mg/kg)	0.04	0.3	0.07	0.08	0.21	0.04 to 0.05	0.25 to 0.3
Intubating dose (mg/kg)	0.1 to 0.2 ^Δ	0.6 to 1 (1.2 with RSII dose) [◇]	0.08 to 0.12	0.2	0.5 to 0.6	0.15 to 0.2 [§]	0.6 to 1.5
Onset time (min)	3 to 4	1 to 2	2 to 3	3 to 4	3 to 5	4 to 6	1
Time to 25% recovery (min)	20 to 35	30 to 50 (60 to 80 with RSII dose)	60 to 120	15 to 20	20 to 35	30 to 60	5 to 10
Elimination half-life (min)							
■ Normal organ function	50 to 60	60 to 100	100 to 130	2 to 2.5	21	23 to 30	<1
■ Renal impairment	Mild increase	100 to 300	Increased ×2	3 to 4	21	Mild increase	<1
■ Hepatic impairment	Significant increase	120 to 400	Increased ×2	3 to 6	21	23 to 30	<1
Intermittent maintenance dose (mg/kg) [¥]	0.01	0.1	0.02	0.1	0.1	0.01	N/A
Starting infusion dose (mcg/kg/min) [‡]	1 to 2	5 to 12	Not recommended	5 to 8	4 to 12	1 to 3	
Elimination route/metabolism	Renal 10 to 50%; hepatic 30 to 50%	Renal 30%; hepatic 70%	Renal 40 to 70%; hepatic 20%	Plasma cholinesterase (70% of succinylcholine rate)	Renal 10%; Hoffman 30%; ester hydrolysis 60%	Hoffman 30%; ester hydrolysis 60%	Butyrylcholinesterase (plasma cholinesterase, pseudocholinesterase)
Active metabolites	3-desacetyl-vecuronium	17-desacetyl-rocuronium (minimal)	3-OH-pancuronium; 17-OH-pancuronium	No active metabolites	No active metabolites	No active metabolites	No active metabolites
Side effects	Vagal blockade with large doses	Minimal	Vagal block (tachycardia), catecholamine release	Histamine release	Histamine release; laudanosine and acrylates production	None	Myalgia; bradycardia; asystole in children with repeated doses; dual (phase II, competitive) block; anaphylaxis
Contraindications (other than specific allergy)	None	None	Short surgical procedures (<60 min); not recommended for continuous infusion	Pseudocholinesterase deficiency	Hemodynamically unstable patients due to histamine release	None	High K ⁺ ; MH; muscular dystrophy; children; receptor regulation settings; pseudocholinesterase deficiency
Comments	Not for prolonged ICU administration (myopathy); reversible by sugammadex; elimination half-life halved in late pregnancy; 3-desacetyl metabolite has 60% of the parent compound potency	Pain on injection; easily reversible by sugammadex; elimination half-life prolonged in ICU patient; 17-desacetyl metabolite has 20% activity	Significant accumulation, prone to residual block (3-OH metabolite has 50% activity of pancuronium)	Reversal by cholinesterase inhibitors; mixture of 3 isomers (cis-cis minimal); edrophonium for antagonism more effective during deep block	Organ-independent elimination	Trivial histamine release; minimal plasma laudanosine and acrylate levels	Fastest onset, most reliable NMBA for rapid tracheal intubation

The data in this table are averages obtained from published literature and do not account for other influences such as the use of volatile anesthetics or muscle temperature. For further information refer to UpToDate content on clinical use of neuromuscular blocking agents.

ED₉₅: effective dose to achieve 95% depression of baseline muscle contraction; ICU: intensive care unit; K⁺: potassium; MH: malignant hyperthermia; N/A: data not available; NMBA: neuromuscular blocking agents; RSII: rapid sequence induction and intubation; ST: single twitch.

* Pancuronium is no longer available in the North America, Europe, and most areas around the world.

¶ Mivacurium is no longer available in the United States or Canada. It is available in many other countries.

Δ Vecuronium at a dose of 0.1 to 0.2 mg/kg may be used for RSII in patients for whom succinylcholine is contraindicated and rocuronium is unavailable; onset of paralysis will be delayed compared with those preferred agents.

◇ Some experts use a higher dose, 1.5 mg/kg IV, of rocuronium for rapid sequence induction in the emergency department. Refer to UpToDate content on rapid sequence induction for emergency intubation.

§ Cisatracurium at a dose of 0.4 mg/kg may be used for RSII in patients for whom succinylcholine is contraindicated and rocuronium is unavailable; onset of paralysis will be delayed compared with those preferred agents.

¥ Intermittent dosing titrated to endpoints such as ventilator synchrony may be preferred to continuous infusions in the ICU. Higher intermittent maintenance doses may be used in the ICU to provide a longer duration of action and to avoid continuous infusion.

‡ The starting infusion dose presumes that an intubating dose or a similar loading dose has been given prior to starting the infusion. The infusion dose should be adjusted thereafter based on monitoring.

Adapted from: Brull SJ. Neuromuscular blocking agents. In: Clinical Anesthesia, 8th ed, Barash PG, Cullen BF, Stoelting RK, et al (Eds), Wolters Kluwer, Philadelphia 2017.

Graphic 114271 Version 9.0

Table 6: Suggested management of neuromuscular blockade according to monitoring

Modality/response*	Interpretation/management	Comments
Intubating conditions		
TOFC 0 TOFR 0	Adequate conditions likely	During onset of neuromuscular block, diaphragm and vocal cords become paralyzed prior to the adductor pollicis muscle. Diaphragm and vocal cords also recover prior to adductor pollicis.
TOFC ≥ 1 TOFR ≥ 0	Adequate conditions unlikely	Profound block is required for optimal intubating conditions.
Intraoperative conditions		
TOFC 0, PTC 1 to 5	Deep blockade; return of diaphragm movements possible, twitch may return in 5 to 15 minutes, may have twitch response at face with PNS	
TOFC 1 to 3	Moderate blockade; sufficient for most procedures	Deeper levels of block may be warranted in specific settings in which any unexpected movement must be avoided (eg, during delicate dissection under the microscope during craniotomy).
TOFC 4 $\pm$ fade	Shallow or minimal block; additional NMBA may be needed	
Management of recovery and reversal*		
TOFC 0, PTC 0	Complete or profound blockade; wait or administer sugammadex 16 mg/kg IV	- Sugammadex can be used to reverse only rocuronium or vecuronium. - Neostigmine can be used to reverse any nondepolarizing NMBA. - Neostigmine should only be used if the TOFC is 4.
TOFC 0, PTC 1 or 2	Deep blockade; wait or administer sugammadex 4 mg/kg IV	
TOFC 1	Moderate blockade; wait or administer sugammadex 4 mg/kg IV	
TOFC 2 or 3	Moderate blockade; wait or administer sugammadex 2 mg/kg IV	
TOFC 4 with fade TOFR < 0.4	Shallow blockade; wait or sugammadex 2 mg/kg IV or neostigmine 20 to 50 mcg/kg IV	
TOFC 4 without fade TOFR 0.4 to 0.9	Minimal blockade; wait or sugammadex 2 mg/kg IV or neostigmine 15 to 30 mcg/kg IV	
TOFR ≥ 0.9	Fully reversed; no need to administer reversal prior to extubation	

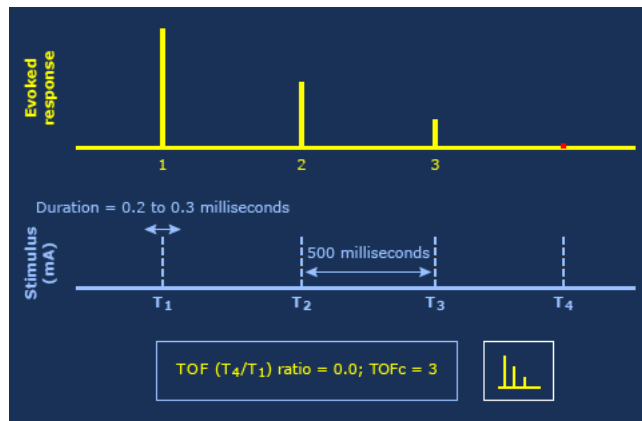
This table shows management according to qualitative and/or quantitative neuromuscular monitoring. Qualitative (subjective) monitoring refers to visual or tactile assessment of adduction of the thumb in response to ulnar nerve stimulation and the resulting contraction of the adductor pollicis muscle. Quantitative (objective) monitors measure the response of the muscle to neurostimulation and provide a ratio of the fourth to the first twitch in the train of four stimuli. Quantitative monitors should be used whenever they are available. For further information, refer to UpToDate content on the clinical use of NMBAs and monitoring neuromuscular blockade.

TOFC: train of four count; TOFR: train of four ratio; NMBA: neuromuscular blocking agent; PTC: post tetanic count; PNS: peripheral nerve stimulator.

* Train of four stimulation consists of four successive supramaximal stimuli using a peripheral nerve stimulator. After administration of a full dose of nondepolarizing NMBA, responses show fade, or progressively decreasing amplitude of the responses; with increasing block the number of responses to a train of four stimuli decreases. When the train of four count is zero, further information on the depth of block can be obtained by applying tetanic stimulation, followed by a series of single twitches for 20 seconds. The post tetanic count (PTC) refers to the number of twitches apparent in this series; increasing PTC correlates with less profound block.

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Figure 1: Train-of-four count (TOFC) - Moderate block

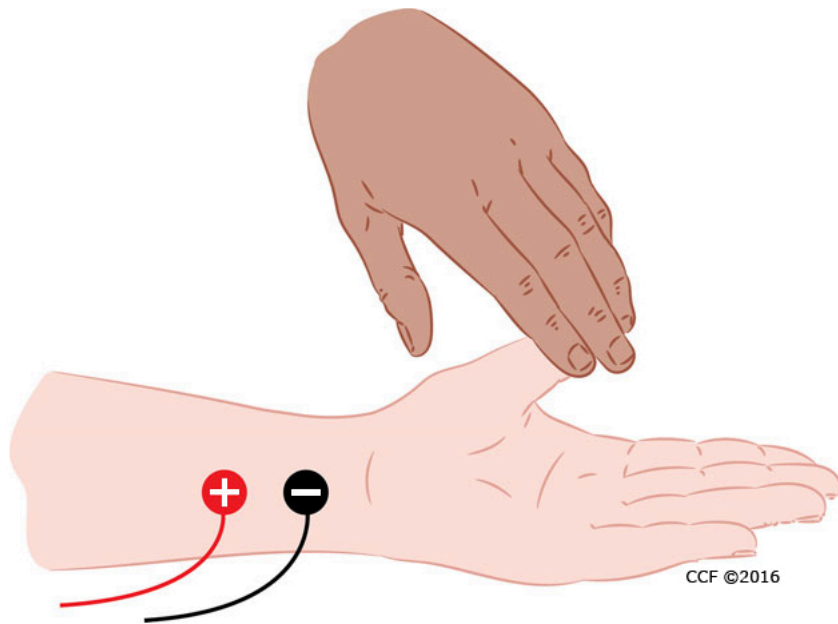


Pattern of TOF stimulation (lower panel) and the evoked muscle response during partial (moderate) neuromuscular block (upper panel). Note there is no muscle response to the fourth TOF stimulus. For further information, refer to UpToDate content on monitoring neuromuscular blockade.

mA: milliamperes; T₁: first twitch in sequence; T₂: second twitch in sequence; T₃: third twitch in sequence; T₄: fourth twitch in sequence; TOF: train-of-four; TOFC: train-of-four count.

Graphic 114139 Version 3.0

Figure 2: Subjective evaluation of neuromuscular responses at the adductor pollicis



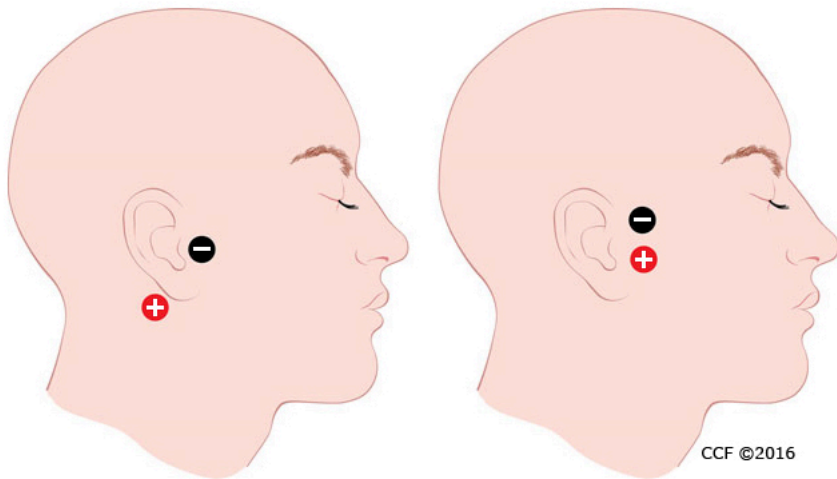
Subjective (tactile) evaluation of neuromuscular responses at the adductor pollicis (thumb) muscle in response to ulnar nerve stimulation. Note the negative (black) electrode is placed distally.

Reference:

1. Naguib M, Brull SJ, Johnson KB. Conceptual and technical insights into the basis of neuromuscular monitoring. *Anaesthesia* 2017; 72 Suppl 1:16. Reprinted with permission, Cleveland Clinic Center for Medical Art & Photography ©2016-2017. All Rights Reserved.

Graphic 114145 Version 1.0

Figure 3: Placement of stimulating electrodes for monitoring of eye



Suggested placement of stimulating electrodes for monitoring of the eye (orbicularis oculi, corrugator supercilii) muscles. Given the course of the facial nerve, note that the positive (red) electrode is always proximal and the negative (black) electrode always distal.

Reference:

1. Naguib M, Brull SJ, Johnson KB. Conceptual and technical insights into the basis of neuromuscular monitoring. *Anaesthesia* 2017; 72 Suppl 1:16. Reprinted with permission, Cleveland Clinic Center for Medical Art & Photography ©2016-2017. All Rights Reserved.

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Table 7: Recommendations from the ASA guidelines for neuromuscular blockade

1. When neuromuscular blocking drugs are administered, we recommend against clinical assessment alone to avoid residual neuromuscular blockade, due to the insensitivity of the assessment.
2. We recommend quantitative monitoring over qualitative assessment to avoid residual neuromuscular blockade.
3. When using quantitative monitoring, we recommend confirming a train-of-four ratio greater than or equal to 0.9 before extubation.
4. We recommend using the adductor pollicis muscle for neuromuscular monitoring.
5. We recommend against using eye muscles for neuromuscular monitoring.
6. We recommend sugammadex over neostigmine at deep, moderate, and shallow depths of neuromuscular blockade induced by rocuronium or vecuronium, to avoid residual neuromuscular blockade.*
7. We suggest neostigmine as a reasonable alternative to sugammadex at minimal depth of neuromuscular blockade.
8. To avoid residual neuromuscular blockade when atracurium or cisatracurium are administered and qualitative assessment is used, we suggest antagonism with neostigmine at minimal neuromuscular blockade depth. In the absence of quantitative monitoring, at least 10 minutes should elapse from antagonism to extubation. When quantitative monitoring is utilized, extubation can be done as soon as a train-of-four ratio greater than or equal to 0.9 is confirmed before extubation.

* Deep: posttetanic count greater than or equal to 1 and train-of-four count 0; moderate: train-of-four count 1 to 3; shallow: train-of-four count 4 and train-of-four ratio less than 0.4; minimal: train-of-four ratio 0.4 to less than 0.9.

Adapted with permission from Wolters Kluwer Health, Inc.: Thilen SR, Weigel WA, Todd MM, et al. 2023 American Society of Anesthesiologists practice guidelines for monitoring and antagonism of neuromuscular blockade: A report by the American Society of Anesthesiologists Task Force on Neuromuscular Blockade. Anesthesiology 2023; 138(1):13-41. Copyright © 2023 American Society of Anesthesiologists. pubs.asahq.org/anesthesiology.

Table 8: Classification of depth of nondepolarizing neuromuscular blockade based on subjective and objective criteria

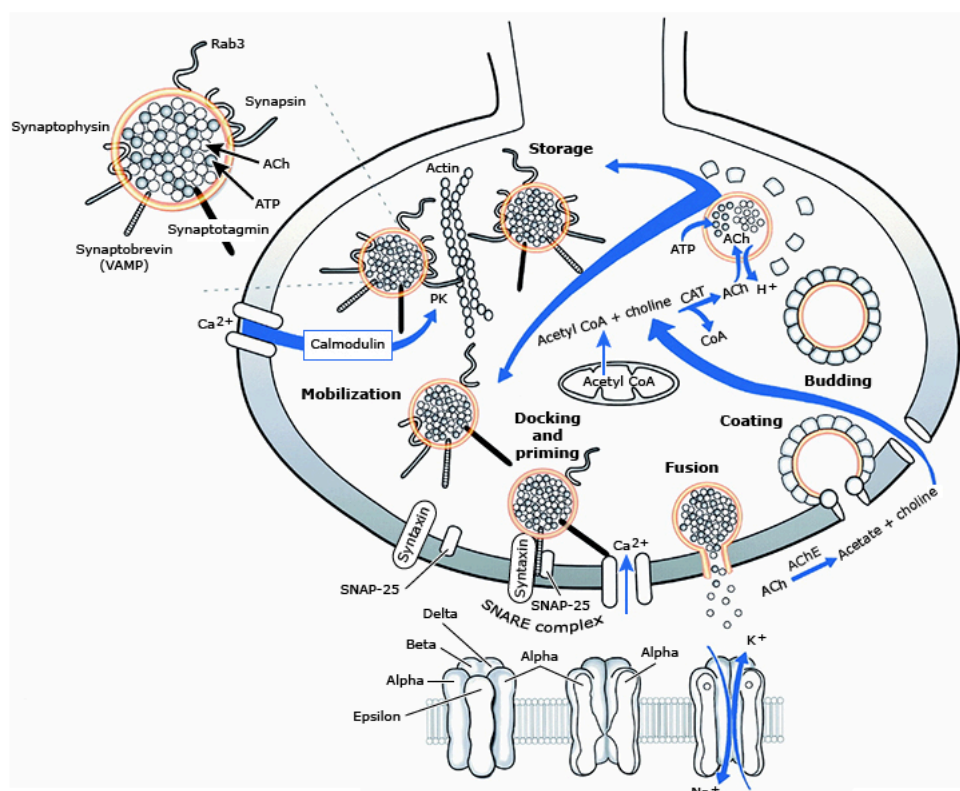
Depth of block	Posttetanic count (PTC)	Train-of-four count (TOFC)	Subjective train-of-four (TOF) ratio	Measured (actual) train-of-four (TOF) ratio
Intense (profound) block	0	0	0	0
Deep block	≥1	0	0	0
Moderate block	NA	1 to 3	0	0
Light (shallow) block	NA	4	Fade present	0.1 to 0.4
Minimal block	NA	4	No fade	>0.4 but <0.9
Full recovery	NA	4	No fade	≥0.9 to 1.0

NA: data not available.

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Figure 4: Neuromuscular junction



The synaptic vesicle exocytosis-endocytosis cycle. After an action potential and Ca^{2+} influx, phosphorylation of synapsin is activated by calcium-calmodulin activated protein kinases I and II. This results in the mobilization of SVs from the cytomatrix toward the plasma membrane. The formation of the SNARE complex is an essential step for the docking process. After fusion of SVs with the presynaptic plasma membrane, ACh is released into the synaptic cleft. Some of the released acetylcholine molecules bind to the nAChRs on the postsynaptic membrane, while the rest is rapidly hydrolyzed by the AChE present in the synaptic cleft to choline and acetate. Choline is recycled into the terminal by a high-affinity uptake system, making it available for the resynthesis of acetylcholine. Exocytosis is followed by endocytosis in a process dependent on the formation of a clathrin coat and of action of dynamin. After recovering of SV membrane, the coated vesicle uncoats and another cycle starts again.

Acetyl CoA: acetylcoenzyme A; CAT: choline acetyltransferase; PK: protein kinase; SVs: synaptic vesicles; ACh: acetylcholine; nAChRs: nicotinic acetylcholine receptors; AChE: acetylcholinesterase.

From: Naguib M, Flood P, McArdle JJ, Brenner HR. Advances in neurobiology of the neuromuscular junction: implications for the anesthesiologist. *Anesthesiology* 2002; 96:202. Copyright © 2002 American Society of Anesthesiologists. Reproduced with permission from Lippincott Williams & Wilkins. Unauthorized reproduction of this material is prohibited.

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Contributor Disclosures

Johnathan Ross Renew, MD Grant/Research/Clinical Trial Support: Merck [Neuromuscular blockade]. All of the relevant financial relationships listed have been mitigated. **Girish P Joshi, MB, BS, MD, FFARCSI, FASA** No relevant financial relationship(s) with ineligible companies to disclose. **Nancy A Nussmeier, MD, FAHA** No relevant financial relationship(s) with ineligible companies to disclose.

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